Sumitomo Bakelite Co. (4203.T)

Leader in AI thermal management and LEO satellite advanced packaging; M&A expands the growth story for key EMC player; initiate at Buy

4203.T	12m Price Target: ¥11,760	Price: ¥7,839	Upside: 50.0%

BUY

Atsushi Ikeda
+81(3)4587-9940 | atsushi.ikeda@gs.com
Goldman Sachs Japan Co., Ltd.

Yuri Izumikawa
+81(3)4587-3643 | yuri.x.izumikawa@gs.com
Goldman Sachs Japan Co., Ltd.

We initiate coverage of Sumitomo Bakelite with a Buy rating and a 12-month target price of JPY11,760, implying 50% upside. The company holds the world's no. 2 position in semiconductor back-end materials and is the global leader in semiconductor encapsulation materials (EMC; epoxy molding compounds), with over 40% market share. Supported by the expansion of generative AI and the evolution of advanced packaging technologies, Sumitomo Bakelite's "niche top" strategy—underpinned by high entry barriers and strong pricing power—is bearing fruit. We believe the company has entered a phase of robust profit growth.

While the stock has performed strongly over the past six months, we believe further re-rating potential remains. In our view, the market has yet to fully reflect the company's earnings growth opportunities stemming from rising thermal-management demand in AI servers and its emerging exposure to satellite communications applications. Our FY3/27E-FY3/29E operating profit forecasts are 15%/26%/38% above Bloomberg consensus.

Thermal management leader benefiting from AI server adoption and rising heat generation

As AI semiconductors become increasingly widespread, power density and performance continue to rise, pushing thermal challenges to new extremes. Advanced heat dissipation solutions are becoming a critical determinant of packaging performance. Leveraging its proprietary inorganic filler formulation and interface-control technologies, Sumitomo Bakelite has established leading expertise in high thermal conductivity materials, securing what we view as a near de facto standard position in applications such as GPU boards and power devices used in AI data centers.

The rapid growth in AI-related electricity demand is driving a favorable mix shift toward premium high-thermal-conductivity grades, which carry attractive incremental margins. As a result, we

Key Data

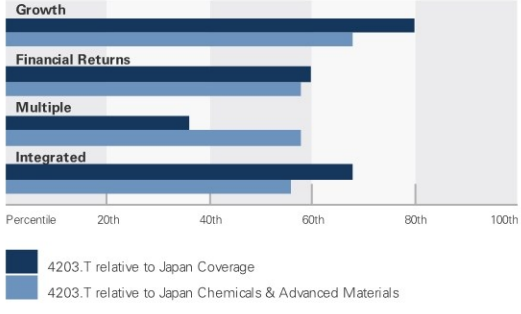
Market cap: ¥687.8bn / $4.4bn
Enterprise value: ¥590.9bn / $3.7bn
3m ADTV: ¥6.4bn / $40.9mn
Japan
Japan Chemicals & Advanced Materials
M&A Rank: 3
Leases incl. in net debt & EV?: Yes

GS Forecast

	3/26	3/27E	3/28E	3/29E
Revenue (¥ bn)	319.9	376.6	424.0	465.4
Op. profit (¥ bn)	35.5	55.7	71.4	85.4
Op. profit CoE (¥ bn)	–	37.5	–	–
EPS (¥)	319.5	465.4	591.7	703.2
P/E (X)	14.6	16.8	13.2	11.1
P/B (X)	1.2	1.8	1.6	1.4
Dividend yield (%)	2.2	2.0	2.8	3.6
N debt/EBITDA (ex lease,X)	(2.0)	(1.4)	(1.2)	(1.3)
CROCI (%)	7.1	9.7	11.2	12.2

	6/26	9/26E	12/26E	3/27E
EPS (¥)	124.6	106.1	118.9	115.8

GS Factor Profile



Source: Company data, Goldman Sachs Research estimates.
See disclosures for details.

Goldman Sachs does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision. For Reg AC certification and other important disclosures, see the Disclosure Appendix, or go to www.gs.com/research/hedge.html. Analysts employed by non-US affiliates are not registered/qualified as research analysts with FINRA in the U.S.

BUY

Sumitomo Bakelite Co. (4203.T)

Rating since Oct 27, 2026

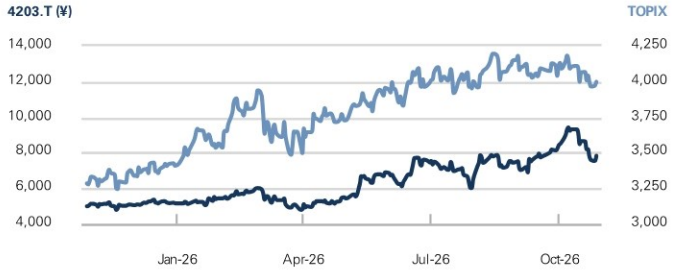
Ratios & Valuation

	3/26	3/27E	3/28E	3/29E
P/E (X)	14.6	16.8	13.2	11.1
P/B (X)	1.2	1.8	1.6	1.4
FCF yield (%)	3.8	2.0	4.1	6.7
EV/EBITDAR (X)	6.3	8.3	6.6	5.4
EV/EBITDA (excl. leases) (X)	6.3	8.3	6.6	5.4
CROCI (%)	7.1	9.7	11.2	12.2
ROE (%)	8.8	11.2	12.7	13.6
Net debt/equity (%)	(28.8)	(26.0)	(25.4)	(27.2)
Net debt/equity (excl. leases) (%)	(28.8)	(26.0)	(25.4)	(27.2)
Interest cover (X)	89.8	206.3	793.3	854.0
Days inventory outst, sales	74.6	70.5	69.9	69.4
Receivable days	63.4	61.1	62.3	63.1
Days payable outstanding	74.5	72.4	72.8	74.0
DuPont ROE (%)	8.0	10.5	11.9	12.7
Turnover (X)	0.7	0.7	0.7	0.7
Leverage (X)	1.4	1.3	1.3	1.3
Gross cash invested (ex cash) (¥)	539.5	583.2	625.2	660.3
Average capital employed (¥)	234.1	268.7	306.0	339.1
BVPS (¥)	3,996.6	4,431.5	4,951.6	5,544.7

Growth & Margins (%)

	3/26	3/27E	3/28E	3/29E
Total revenue growth	5.0	17.7	12.6	9.8
EBITDA growth	28.3	43.2	24.7	17.2
EPS growth	52.9	45.6	27.1	18.9
DPS growth	10.5	52.4	37.5	27.3
EBIT margin	11.1	14.8	16.8	18.3
EBITDA margin	15.5	18.9	20.9	22.4
Net income margin	8.8	10.8	12.2	13.3

Price Performance



	3m	6m	12m
Absolute	18.0%	48.2%	50.5%
Rel. to the TOPIX	19.9%	38.4%	25.1%

Source: FactSet. Price as of 27 Oct 2026 close.

Income Statement (¥ bn)

	3/26	3/27E	3/28E	3/29E
Total revenue	319.9	376.6	424.0	465.4
Cost of goods sold	(219.8)	(246.7)	(274.0)	(293.6)
SG&A	(65.6)	(75.3)	(79.7)	(87.5)
R&D	–	–	–	–
Other operating inc./(exp.)	1.0	1.1	1.1	1.1
EBITDA	**49.7**	**71.2**	**88.8**	**104.1**
Depreciation & amortization	(14.3)	(15.5)	(17.4)	(18.7)
EBIT	**35.5**	**55.7**	**71.4**	**85.4**
Net interest inc./(exp.)	3.4	3.5	3.7	3.9
Income/(loss) from associates	–	–	–	–
Pre-tax profit	**38.8**	**59.2**	**75.1**	**89.3**
Provision for taxes	(10.5)	(17.8)	(22.5)	(26.8)
Minority interest	(0.4)	(0.6)	(0.7)	(0.8)
Preferred dividends	–	–	–	–
Net inc. (pre-exceptionals)	**28.0**	**40.8**	**51.9**	**61.7**
Post-tax exceptionals	–	–	–	–
Net inc. (post-exceptionals)	**28.0**	**40.8**	**51.9**	**61.7**
EPS (basic, pre-except) (¥)	**319.5**	**465.4**	**591.7**	**703.2**
EPS (diluted, pre-except) (¥)	**319.5**	**465.4**	**591.7**	**703.2**
EPS (basic, post-except) (¥)	**319.5**	**465.4**	**591.7**	**703.2**
EPS (diluted, post-except) (¥)	**319.5**	**465.4**	**591.7**	**703.2**
DPS (¥)	105.0	160.0	220.0	280.0
Div. payout ratio (%)	32.9	34.4	37.2	39.8

Balance Sheet (¥ bn)

	3/26	3/27E	3/28E	3/29E
Cash & cash equivalents	128.4	121.0	130.3	152.5
Accounts receivable	57.9	68.1	76.7	84.2
Inventory	68.6	77.0	85.5	91.6
Other current assets	12.1	12.7	13.4	14.0
Total current assets	**267.0**	**278.8**	**305.9**	**342.3**
Net PP&E	141.4	155.9	168.5	174.8
Net intangibles	5.7	5.7	5.7	5.7
Total investments	0.0	0.0	0.0	0.0
Other long-term assets	70.1	80.2	91.9	105.3
Total assets	**484.2**	**520.7**	**572.0**	**628.2**
Accounts payable	46.1	51.8	57.5	61.6
Short-term debt	20.1	20.1	20.1	20.1
Short-term lease liabilities	–	–	–	–
Other current liabilities	24.8	24.8	24.8	24.8
Total current liabilities	**91.0**	**96.7**	**102.4**	**106.5**
Long-term debt	7.3	–	–	–
Long-term lease liabilities	–	–	–	–
Other long-term liabilities	35.2	35.2	35.2	35.2
Total long-term liabilities	**42.5**	**35.2**	**35.2**	**35.2**
Total liabilities	**133.5**	**131.9**	**137.6**	**141.7**
Preferred shares	–	–	–	–
Total common equity	**347.2**	**384.7**	**429.7**	**480.9**
Minority interest	**3.5**	**4.1**	**4.8**	**5.6**
Total liabilities & equity	**484.2**	**520.7**	**572.0**	**628.2**
Net debt, adjusted	(101.1)	(100.9)	(110.3)	(132.4)

Cash Flow (¥ bn)

	3/26	3/27E	3/28E	3/29E
Net income	28.0	40.8	51.9	61.7
D&A add-back	14.3	15.5	17.4	18.7
Minority interest add-back	0.4	0.6	0.7	0.8
Net (inc)/dec working capital	(4.1)	(13.0)	(11.4)	(9.5)
Other operating cash flow	(3.5)	–	–	–
Cash flow from operations	**35.0**	**43.9**	**58.6**	**71.7**
Capital expenditures	(19.3)	(30.0)	(30.0)	(25.0)
Acquisitions	–	–	–	–
Divestitures	–	–	–	–
Others	11.3	–	–	–
Cash flow from investing	**(7.9)**	**(30.0)**	**(30.0)**	**(25.0)**
Repayment of lease liabilities	–	–	–	–
Dividends paid (common & pref)	(9.2)	(14.0)	(19.3)	(24.6)
Inc/(dec) in debt	(4.7)	(7.3)	–	–
Other financing cash flows	9.7	0.0	0.0	0.0
Cash flow from financing	**(4.2)**	**(21.3)**	**(19.3)**	**(24.6)**
Total cash flow	**22.8**	**(7.4)**	**9.3**	**22.1**
Free cash flow	15.7	13.9	28.6	46.7

Source: Company data, Goldman Sachs Research estimates.

forecast a 43% business profit CAGR for the mainstay Semiconductor Materials segment between FY3/26 and FY3/29.

Developments in advanced packaging and LEO satellite communications, supported by synergies from disciplined M&A

The company has secured a long-term supply agreement with a major US aerospace company for advanced granular encapsulant materials used in large-panel panel-level packaging (PLP), positioning itself within the packaging ecosystem for communications and AI chips used in emerging LEO (low-earth-orbit) satellite communication infrastructure. The company also continues to expand production capacity for molded underfill (MUF) materials targeting memory and AI-PC applications while bringing next-generation liquid encapsulants to market. We also see potential longer term growth in encapsulation materials for actuators used in physical AI applications. Sumitomo Bakelite is due to complete the acquisition of Kyocera's chemical business at a purchase price of c.JPY30 bn, with closing expected by the end of October 2026. We expect the transaction to drive M&A synergies, expanding the company's global EMC market share to more than 45% while strengthening its ability to offer comprehensive backend material solutions through a broader product portfolio.

Portfolio transformation and shareholder returns supported by strong financials: Portfolio transformation is also progressing, supported by growth in the aerospace business within High-Performance Plastics and the emerging potential of cyclo-olefin polymer COPLUS for optical transceiver applications. With support from a strong balance sheet including net cash of over JPY100 bn as of end-FY3/26, we forecast DPS to increase to JPY280 by FY3/29E (implying a payout ratio of c.40%). We see profit growth prospects as attractive, based on our outlook for a 35% consolidated business profit CAGR between FY3/26 and FY3/29.

Within this report we also discuss the **key risks** to our investment thesis and target price: Weaker-than-expected AI server and advanced semiconductor demand; intensifying competition and market-share erosion in EMC, and yen appreciation against USD.

Business Summary / Outlook

Sumitomo Bakelite is a global functional materials leader characterized by its “niche top” business model. Leveraging proprietary formulation expertise and high thermal conductivity technologies developed through its legacy synthetic resin businesses, the company has established itself as the world’s no. 2 supplier of semiconductor back-end materials and the global leader in EMC with more than 40% global market share (set to rise to over 45% following the Kyocera chemicals business integration).

The company operates through three main business segments:

- **Semiconductor Materials**, the primary growth engine
- **High-Performance Plastics**, providing stable earnings and portfolio diversification
- **Quality of Life (QOL) Products**, a stable cash-generating business

Assuming a 43% business profit CAGR in Semiconductor Materials between FY3/26 and FY3/29, coupled with steady growth in High-Performance Plastics and QOL, we forecast a consolidated business profit CAGR of c.35% over the same period.

- **Semiconductor Materials segment:**

  The segment comprises epoxy molding compounds (EMC), photosensitive dielectric materials, die bonding materials, and package substrate materials. As AI semiconductors become increasingly powerful and densely integrated, thermal management has become a critical challenge. Leveraging proprietary high thermal conductivity technology, Sumitomo Bakelite has secured a leading position in thermal management solutions, particularly for power-device applications used in AI servers. The company’s key focus areas span advanced packages, memory, and power devices.

  The company has also secured a long-term agreement with a major US aerospace company for communications and AI-chip packaging used in large-panel panel-level packaging (PLP) and LEO satellite infrastructure. To address strong demand, management is expanding capacity for MUF and advanced granular materials in Suzhou and Singapore while advancing next-generation liquid encapsulants for HBM and advanced packaging applications.

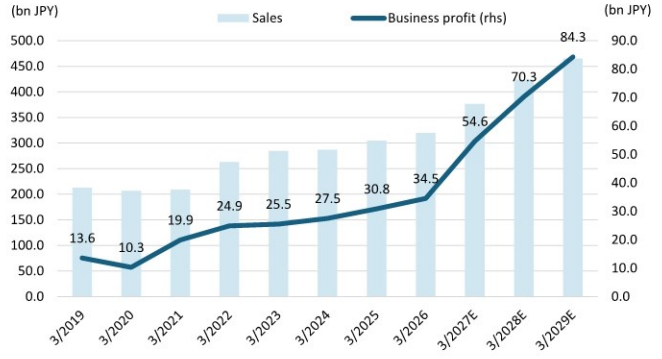
  Further upside could come from the planned integration of Kyocera’s chemicals business (from FY3/27 Q3 onward), which will add memory-customer exposure and high-thermal-conductivity silver sintering paste technologies. We expect the combination to strengthen customer solution capabilities across the back-end materials value chain while potentially generating meaningful M&A synergies.

- **High-Performance Plastics and QOL Products:**

  The High-Performance Plastics segment is built around phenolic molding compounds and industrial resins. We expect growth to be driven by the aerospace business, supported by ongoing structural reforms, rising MRO demand, and increasing adoption in next-generation aircraft such as the A350F. We also see potential from new products such as COPLUS, a high heat-resistant cycloolefin polymer. Meanwhile, we look for the QOL segment to continue generating stable

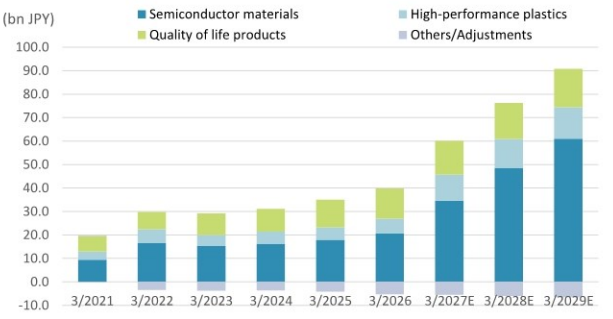
profits, supported by products including cover tapes and dicing tapes for semiconductor transportation and processing, as well as high-performance polycarbonate sheets.

Exhibit 1: Company-wide sales and business profit trends
Consolidated business profit CAGR of +35% (FY3/26-FY3/29, GSE)



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 2: Business profit by segment
Rapid semiconductor materials business profit growth: +43% CAGR (FY3/26-FY3/29, GSE)



Source: Company data, Goldman Sachs Global Investment Research

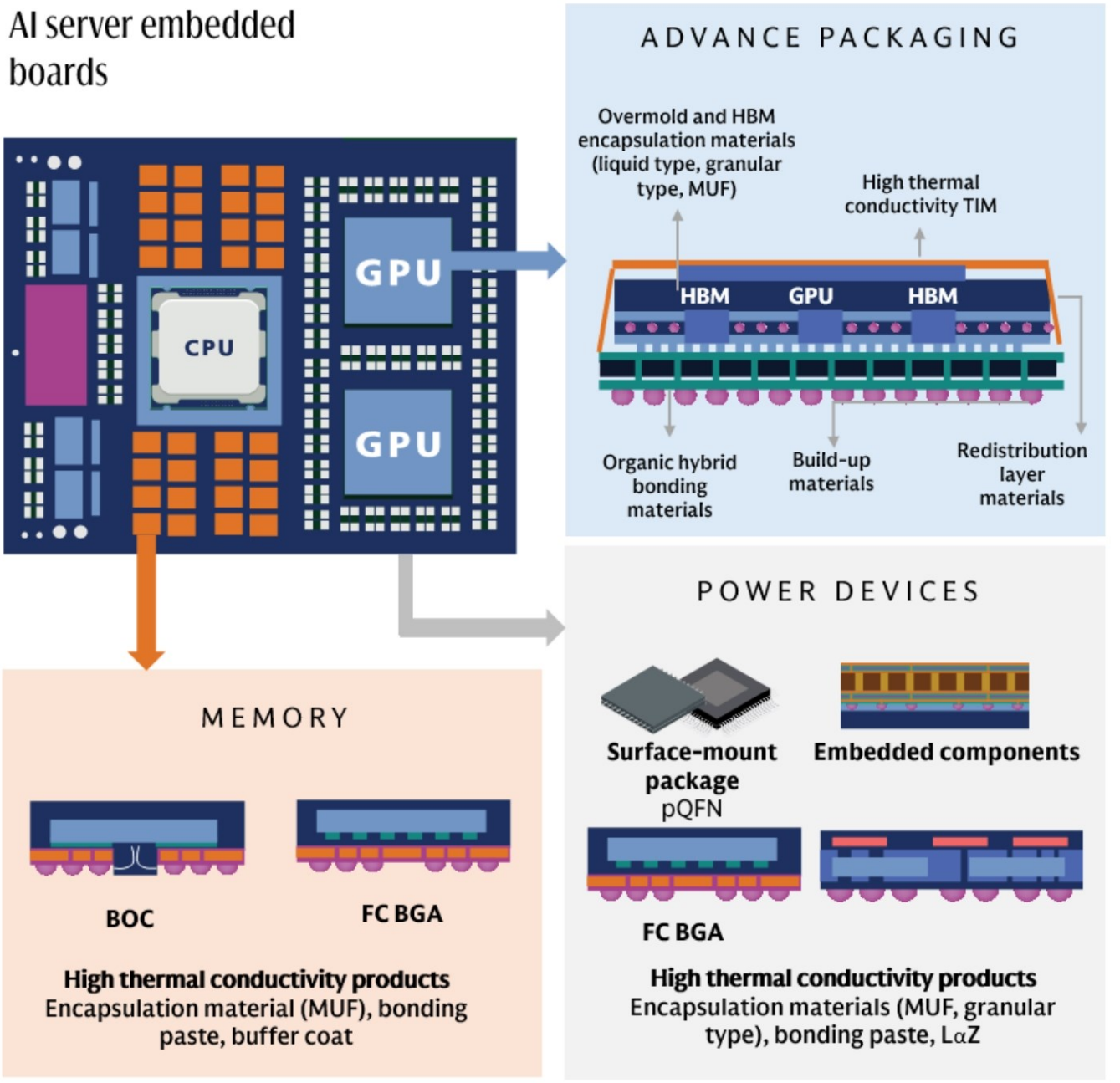
Exhibit 3: Earnings summary

(Bn JPY)				GSE	GSE	GSE	CoE						GSE	GSE
Sales	3/2024	3/2025	3/2026	3/2027E	3/2028E	3/2029E	3/2027E	3/2026 1Q	3/2026 2Q	3/2026 3Q	3/2026 4Q	3/2027 1Q	3/2027 2Q	3/2027 3Q
Semiconductor materials	82.9	91.3	106.4	146.5	182.5	213.0	117.5	24.4	26.8	27.4	27.8	34.2	35.3	40.4
High-performance plastics	101.4	105.5	105.5	116.9	123.8	131.2	109.5	26.0	25.1	26.1	28.2	31.9	29.2	29.9
Quality of life products	102.2	107.2	107.2	112.5	117.0	120.5	109.0	27.1	26.7	27.6	25.8	28.9	28.3	29.3
Others/Adjustments	0.8	0.8	0.8	0.7	0.7	0.7	0.0	0.2	0.2	0.2	0.2	0.2	0.2	0.2
Total	**287.3**	**304.8**	**319.9**	**376.6**	**424.0**	**465.4**	**337.0**	**77.7**	**78.9**	**81.3**	**82.0**	**95.2**	**93.0**	**99.7**
Business Profit														
Semiconductor materials	16.1	18.0	20.7	34.6	48.5	61.0	23.5	4.8	5.4	5.3	5.3	8.0	8.2	9.8
High-performance plastics	5.3	5.3	6.2	11.1	12.4	13.4	7.0	1.7	0.8	1.6	2.0	3.2	2.8	2.7
Quality of life products	9.7	11.8	12.9	14.4	15.4	16.4	13.0	3.7	3.0	3.3	2.9	3.8	3.7	3.8
Others/Adjustments	-3.7	-4.2	-5.4	-5.5	-6.0	-6.5	0.0	-1.2	-1.1	-1.4	-1.6	-1.4	-1.4	-1.4
Total	**27.5**	**30.8**	**34.5**	**54.6**	**70.3**	**84.3**	**38.0**	**9.0**	**8.1**	**8.7**	**8.6**	**13.6**	**13.3**	**14.9**
Business Profit Margin														
Semiconductor materials	19.5%	19.7%	19.5%	23.6%	26.6%	28.6%	20.0%	19.7%	20.0%	19.2%	19.0%	23.4%	23.2%	24.3%
High-performance plastics	5.2%	5.0%	5.9%	9.5%	10.0%	10.2%	6.4%	6.7%	3.3%	6.3%	7.1%	10.0%	9.6%	9.0%
Quality of life products	9.5%	11.0%	12.0%	12.8%	13.2%	13.6%	11.9%	13.6%	11.3%	11.9%	11.3%	13.1%	13.1%	13.0%
Total	**9.6%**	**10.1%**	**10.8%**	**14.5%**	**16.6%**	**18.1%**	**11.3%**	**11.6%**	**10.3%**	**10.8%**	**10.5%**	**14.3%**	**14.3%**	**14.9%**

Source: Company data, Goldman Sachs Global Investment Research

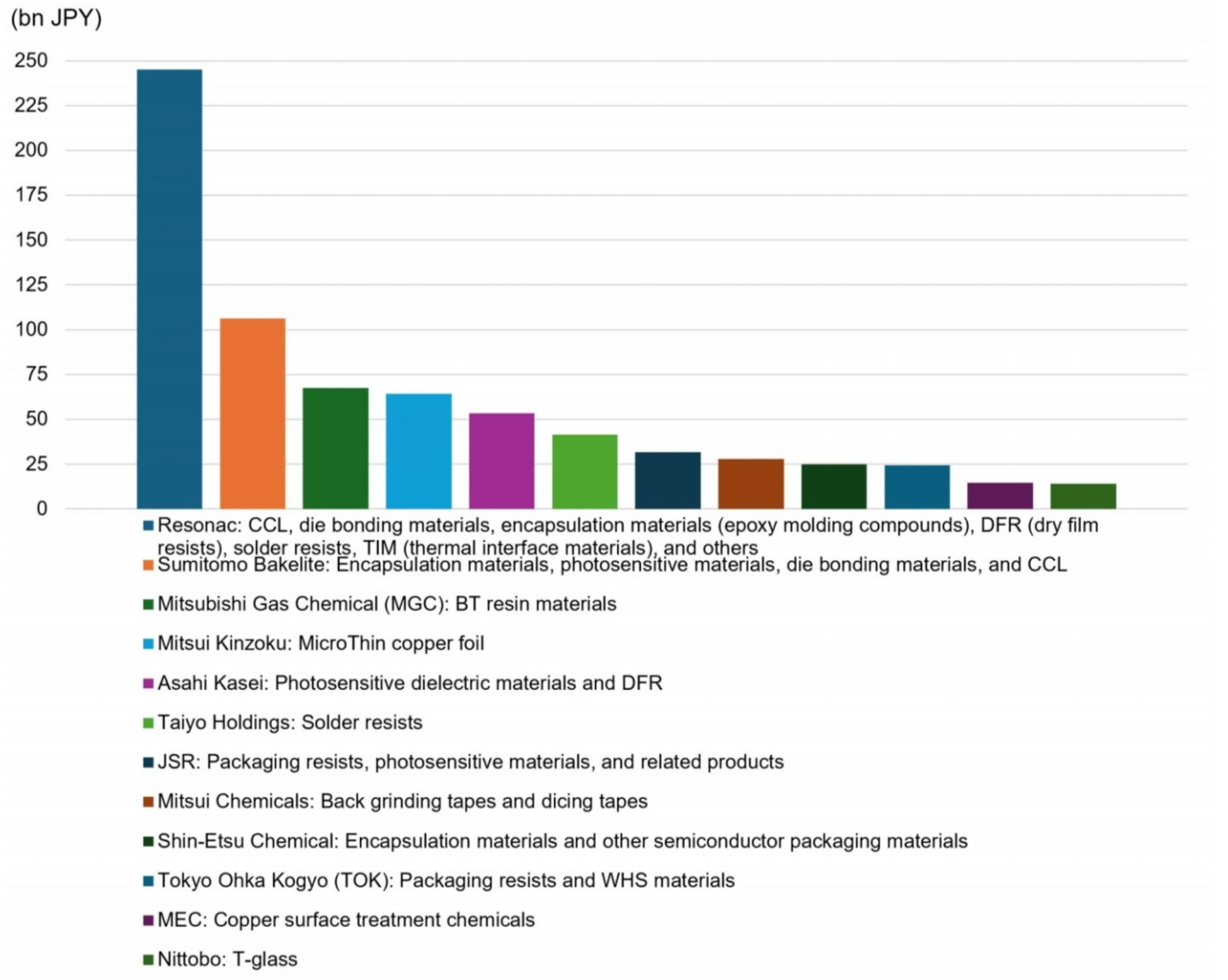
Semiconductor Materials: AI servers, mobility and LEO back-end materials driving growth

Exhibit 4: Sumitomo Bakelite's total solution covering the core of AI servers: GPUs, power systems, and memory



Source: Company data, Goldman Sachs Global Investment Research

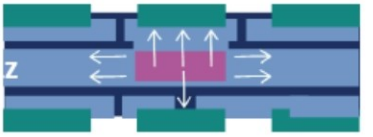
Exhibit 5: Back-end semiconductor materials sales ranking of major Japanese materials players (2025)
Sumitomo Bakelite ranks second, behind Resonac



Source: Company data, Data compiled by Goldman Sachs Global Investment Research

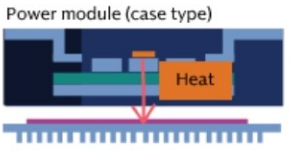
Exhibit 6: An overview of Sumitomo Bakelite's thermal management technologies

- Provide **heat-spreading functionality** directly within the substrate material
- **Reduce dependence on thermal vias** and suppress heat concentration
- Optimize thermal design and circuit design

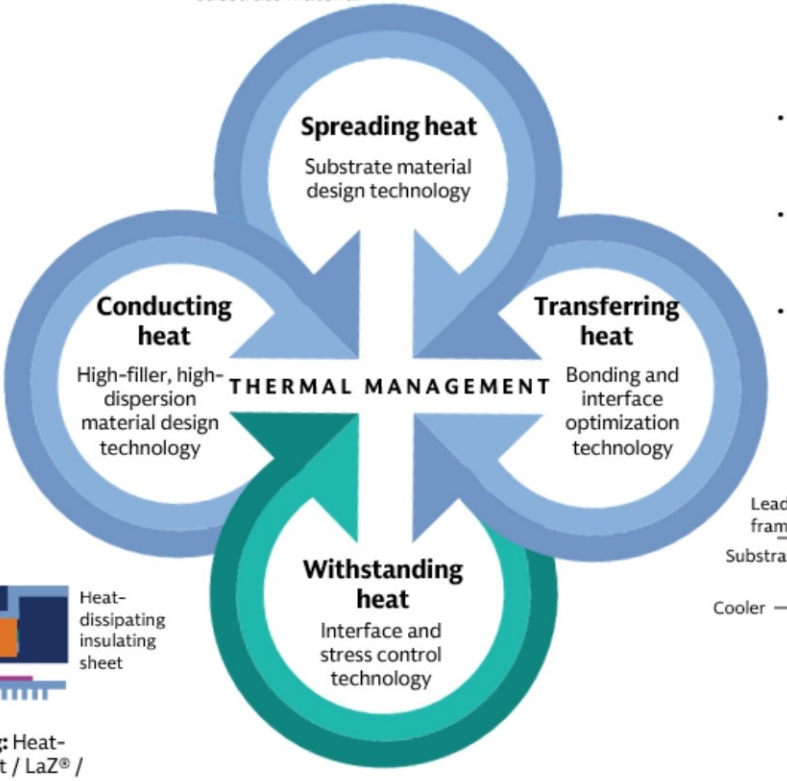


- Improved heat dissipation and reduced # of thermal vias when using a high heat-dissipation substrate material

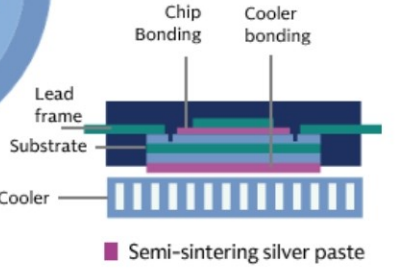
- **Uniformly disperse thermally conductive fillers** at high density within the resin
- Form **thermal conduction pathways while** maintaining insulation/ processability
- Achieve high thermal conductivity and electrical insulation (conflicting requirements)



Related product offering: Heat-dissipating insulating sheet / LaZ® / High thermal conductivity encapsulation material



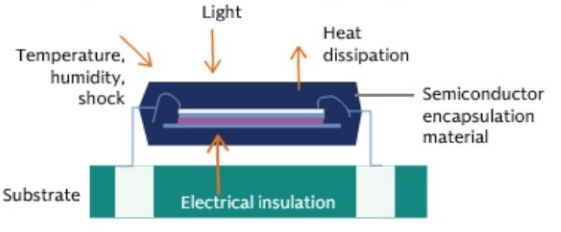
- Reduce interfacial **thermal resistance between the chip and substrate**
- Process design that achieves both **high thermal conductivity and pressureless bonding**
- Improve productivity while efficiently transferring heat to the outside



Related product offering: Semi-sintering silver paste

Contributing to the advancement of ICT and mobility applications

- **Control differences in thermal expansion (CTE mismatch)** between materials caused by temperature changes
- **Relieve stress through interface design** and suppress cracking/warpage
- Ensure long-term reliability even in high-temperature environments



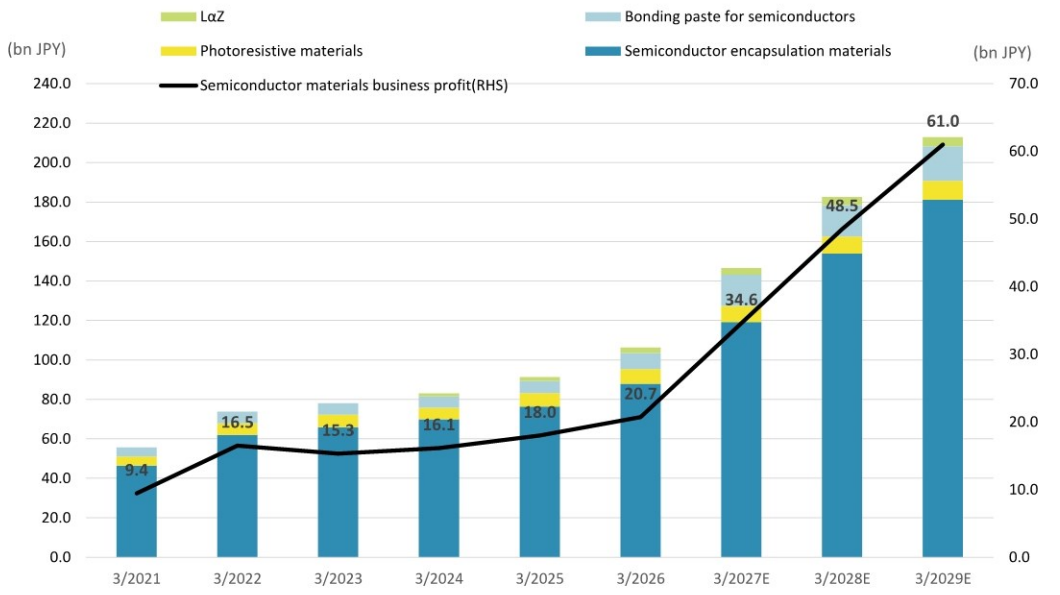
Related product offering: Semiconductor encapsulation material, heat-dissipating insulating sheet

Source: Company data, Goldman Sachs Global Investment Research

Semiconductor Encapsulation Materials (EMC)

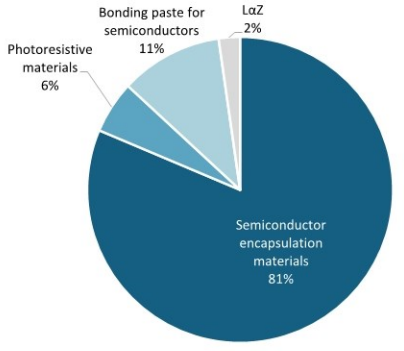
The core of Sumitomo Bakelite's Semiconductor Materials segment is semiconductor encapsulation materials, where the company is the global market leader with more than 40% share, set to rise to over 45% following the integration of Kyocera's chemicals business. We estimate that EMC represented over 80% of Semiconductor Materials segment sales and close to 30% of company-wide sales in FY3/26. Leveraging proprietary resin formulation expertise and advanced thermal management technologies, the company has established a de facto industry standard for addressing the increasing heat generated by higher-power semiconductors. Growth is being driven by the exclusive adoption of its high-thermal-conductivity encapsulants for AI server power devices, advanced granular molding compounds for panel-level packaging (PLP) used in LEO satellite communications, timely capacity expansions for molded underfill (MUF) materials for AI PCs and memory, and the commercialization of next-generation liquid encapsulants. We believe the expansion of these high-value-added products will be a major driver of the company's earnings growth.

Exhibit 7: Semiconductor materials segment: Product sales and segment business profit
Strong business profit growth led by encapsulation materials, backed by steady bonding paste expansion



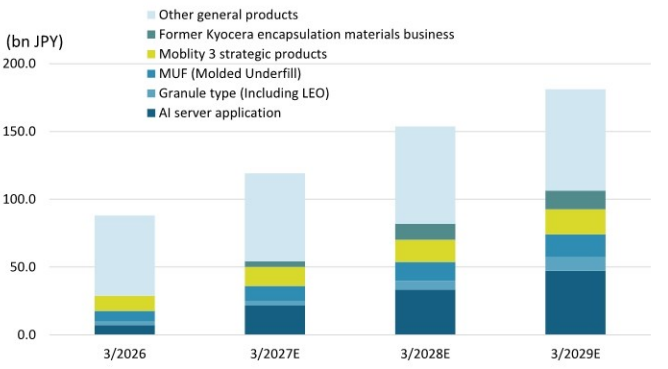
Source: Company data, Goldman Sachs Global Investment Research

Exhibit 8: Semiconductor materials sales breakdown (FY3/27 GSE)



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 9: Encapsulation materials revenue (GSE)
Poised for +27% CAGR (FY3/26-3/29, GSE) on rapid expansion in AI power, MUF, and granular resins



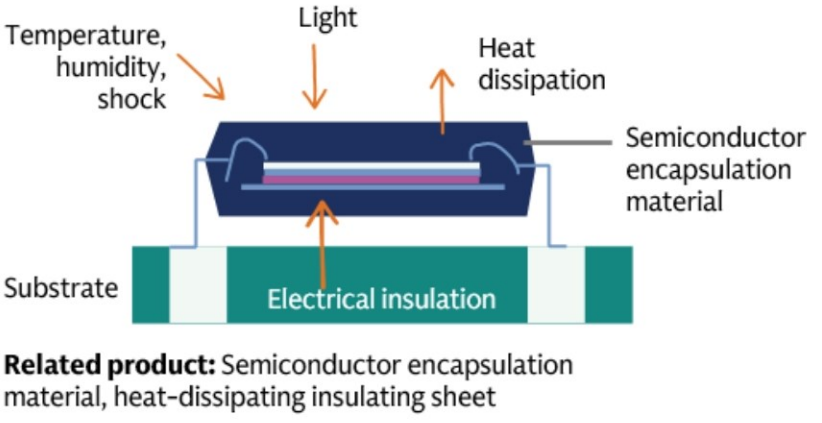
Source: Company data, Goldman Sachs Global Investment Research

Overview of semiconductor encapsulation materials

Semiconductor encapsulation materials, commonly known as epoxy molding compounds (EMC), protect and insulate silicon chips from moisture, ionic contaminants, and mechanical stress while mitigating the thermal stresses generated between chips, substrates, and lead frames.

EMC is a highly sophisticated engineered composite material consisting of an epoxy resin matrix and phenolic curing agent, loaded with spherical silica filler at c.70-90% by weight (typically above 80-90% in high-reliability and high-thermal-conductivity grades). The formulation also incorporates curing accelerators, silane coupling agents, mold-release agents, stress-relief additives such as silicones, pigments, and flame retardants, all of which must be uniformly dispersed.

Exhibit 10: Epoxy molding compounds (EMC) for semiconductors



Source: Company data, Goldman Sachs Global Investment Research

Silica has an extremely low coefficient of thermal expansion (CTE), typically only a fraction of that of the resin matrix. As filler loading increases, the encapsulant exhibits lower thermal expansion and moisture absorption, thereby improving package reliability. However, higher filler loading also causes a sharp increase in melt viscosity, which can lead to molding defects such as wire sweep, residual microvoids, incomplete

filling of narrow gaps, and mold wear. The core of Sumitomo Bakelite's formulation expertise lies in its ability to balance these conflicting requirements through advanced filler surface treatment and resin molecular design.

Semiconductor encapsulation materials can broadly be categorized into four types based on material form and molding process.

- **Transfer Molding Compounds:**

  The mainstream format is the transfer molding compound, supplied in tablet form. Powdered material is compressed into cylindrical tablets, melted by preheating, and injected through runners into molds, enabling multiple packages to be molded simultaneously. This method is widely used across discrete semiconductors, analog and power ICs, automotive ECUs, and power modules due to its high productivity and scalability.

  We estimate that transfer molding compounds accounted for c. 90% of Sumitomo Bakelite's EMC volumes in FY3/26. Beyond standard products, transfer-molded EMC also includes automotive high-reliability grades, high-temperature grades for SiC devices, and alumina-filled high-thermal-conductivity grades used in AI server power devices, particularly for compact leadframe packages directly soldered onto circuit boards.

- **Compression Molding Compounds (Granular EMC)**:

  The second category is compression molding compounds, supplied in granular form. Granular material is distributed across the entire wafer or panel surface and melted before compression molding. Because the resin experiences minimal flow during molding, issues such as wire sweep and incomplete filling are significantly reduced. This makes the technology particularly suitable for thin, large-area, and multi-stack packages while minimizing material loss in runners. Historically, this form of EMC was driven by high-end smartphone applications. The company was also the first in the world to commercialize granular EMC for TSV-based stacked DRAM packages. More recently, shipments have been expanding into LEO satellite panel-level packaging (PLP) applications and LPDDR memory for AI servers. Although current volumes remain small given the recent adoption in LEO applications beginning in 2025, selling prices are high and future growth potential is substantial, particularly in PLP. Reflecting this opportunity, the company announced a roughly threefold expansion of granular EMC production capacity in October 2026 at its Suzhou and Singapore facilities.

- **Molded Underfill (MUF)**:

  The third category, MUF, simultaneously fills the microscopic gaps between flip-chip bumps and encapsulates the chip surface in a single molding step using low-viscosity resin and fine-particle fillers. This reduces process complexity while also improving package warpage control. Applications are concentrated in AI smartphones, AI PCs, and DRAM packages used alongside AI processors, excluding HBM. As chips become larger and generate more heat, demand for high-thermal-conductivity MUF grades is increasing. We estimate MUF represented c. 10% of encapsulation-material sales in FY3/27 Q1. For HBM applications, the company has already begun sample

shipments of a liquid encapsulant capable of simultaneously performing interlayer sealing and overmolding. This could become an important future growth driver if adopted at scale. In October 2026, the company also announced plans to c. double MUF production capacity at its Suzhou and Singapore facilities.

- **Liquid Encapsulants:**

  The fourth category is liquid encapsulants. Although distinguished from solid EMC products, liquid encapsulants utilize similar epoxy-based chemistry and are primarily targeted at wafer-level packaging and HBM applications. The company only recently began customer evaluations of its EME-L Series in April 2026. While volumes remain limited, ASPs are high, making commercialization progress an important area to monitor.

Sumitomo Bakelite's competitive advantages

The essence of thermal-stress design in semiconductor packaging is not simply minimizing the encapsulation material's own thermal expansion, but rather managing the behavior of all surrounding materials to suppress package warpage.

A semiconductor package is composed of materials with vastly different thermal expansion characteristics, including silicon chips, substrates, leadframes, and encapsulation materials. During cooling after molding and repeated solder reflow cycles at c. 260°C, these materials expand and contract at different rates. When the temperature changes by 1°C, silicon expands or contracts by only about 3 μm per meter, while an encapsulation material containing nearly 90% silica is said to expand or contract by 7–12 μm, or roughly 2-4 times as much. Differences in the coefficient of thermal expansion (CTE) between materials can lead to connection failures, delamination, and chip cracking.

Because silica-filled EMC cannot fully match silicon's thermal expansion characteristics, formulations are optimized through a combination of modulus, glass-transition temperature (Tg), and thermal-expansion behavior above and below Tg to minimize package warpage across both ambient and high-temperature conditions. EMC is effectively the only package material whose properties can be extensively tailored through formulation, making it the primary tool for warpage control. Since the optimal formulation depends on chip dimensions and substrate design, EMC qualified for one package cannot simply be transferred to another.

As these design requirements become more stringent, encapsulation materials evolve into highly customized products tied to specific customer device platforms and molding processes. Substitution becomes increasingly difficult, particularly in automotive, power-device, and advanced-packaging applications. Any change to a qualified encapsulation material typically requires extensive requalification testing and renewed approval from end customers such as automotive OEMs and hyperscale cloud operators. In automotive and high-reliability power applications, this process can take years. As a result, switching costs far exceed the cost of the material itself, making it common for an approved encapsulation material to remain in use throughout the entire lifecycle of the device.

Beyond its proprietary, black-box formulation and interface-control know-how, Sumitomo Bakelite's competitive advantages include: a) Extensive qualification and

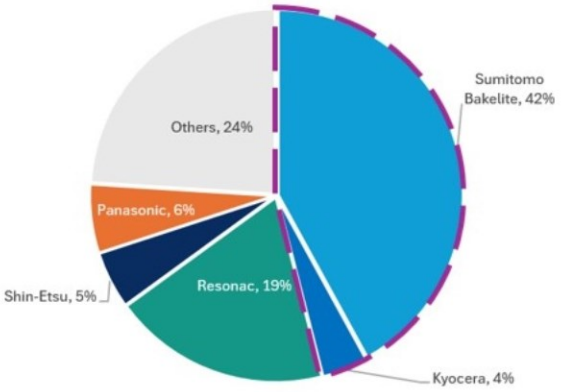
adoption history together with deep customer trust; b) Joint evaluation capabilities through open laboratories in seven countries; and c) An integrated manufacturing, sales, and R&D network spanning Japan, mainland China, Southeast Asia, and Europe

These strengths have enabled Sumitomo Bakelite to establish a c.40% global share of the semiconductor encapsulation market by sales (2025), making it the clear market leader. While Resonac ranks first in overall sales of semiconductor back-end materials (Exhibit 5), Sumitomo Bakelite holds a substantial lead in encapsulation materials alone.

The company is followed by Japanese competitors such as Shin-Etsu Chemical and Panasonic. In mainland China, Hua Hai Cheng Ke's acquisition of Hengshuo Huawei Electronics in March 2025 has likely created a meaningful volume-based position, although exposure appears to be concentrated in commodity applications such as discrete and analog semiconductors.

In addition, Sumitomo Bakelite is scheduled to complete the acquisition of Kyocera's chemicals business (including semiconductor encapsulation materials and bonding pastes) at the end of October 2026. The combination will bring together Kyocera's proprietary high-thermal-conductivity technologies and strong customer relationships in memory semiconductors, which should strengthen Sumitomo Bakelite's competitive position and increase its global EMC market share to over 45%, in our view.

Exhibit 11: Global semiconductor encapsulation materials market share (2026, GSE)



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 12: Sumitomo Bakelite: Encapsulation materials production capacity by region

Country / Region & Site	Key Products & Strategic Role	Capacity Expansion Highlights	Timeline / Status
Japan (Kyushu, Utsunomiya)	Advanced packaging, next-gen SiC power devices	Ongoing modernization and yield optimization as a mother plant	Ongoing
Mainland China (Suzhou)	Domestic China & AI (standard, power devices, MUF, granules)	(1) 5th line addition (+30% total China capacity) (2) New on-site facility (new MUF & granule lines)	(1) Scheduled to begin production in Dec 2028 (2) Scheduled to begin production in FY3/31
Singapore	Southeast Asia OSAT hub, advanced packaging (MUF, granules)	New MUF & granule lines in existing plant (MUF capacity ~2x, granule capacity ~3x)	Scheduled to begin production by the end of FY3/29
Taiwan	Customers in Southeast Asia in addition to Taiwan	Completed new plant in 2024, doubling capacity; fully ramped	Operational
Belgium	European automotive (high-heat xEV inverters, ECU overmolding)	Maintaining stable supply for European automotive Tier-1s	Operational
[M&A] Kyocera Chemical Unit	High thermal conductivity materials and memory-related EMCs (AI servers and memory applications)	Acquisition of the chemical business (approximately JPY 30 billion), adding manufacturing capacity and technology know-how	Late Oct 2026 acquisition close (Expected)

Source: Company data, Goldman Sachs Global Investment Research

Key end markets for semiconductor encapsulation materials

1) AI servers

Within AI servers, Sumitomo Bakelite's encapsulation materials are used primarily not in GPUs themselves, but in power devices located on GPU boards and power-supply units. According to the company, a 1GW-class AI data center contains c.5,000 AI servers. Each server incorporates c.8,000 power devices on GPU boards and 2,000 power devices in power-supply units, resulting in roughly 50 mn power devices per 1GW data center (Exhibit 13).

Power device counts per GPU board increase almost proportionally with power requirements, rising from c.50 devices on legacy 600W boards to 200 devices on current 3,000W boards and an estimated 400 devices on next-generation 6,000W boards (Exhibit 14).

Power devices on GPU boards were encapsulated using standard EMC grades in legacy 600W-class designs. However, beginning with the current generation of 3,000W-class boards, the industry has shifted toward high-thermal-conductivity grades, and next-generation 6,000W-class platforms are also expected to adopt these materials. High-thermal-conductivity EMC achieves significantly higher thermal performance through the use of thermally conductive inorganic fillers, such as spherical alumina, combined with optimized particle-size distributions. While raw material costs are higher than those of standard grades, the greater value-added content supports higher selling prices and incremental margins. As a result, growing AI-server demand should not only drive volume growth but also improve product mix, contributing to higher overall profitability in the encapsulation materials business. Management has indicated that the company's encapsulation materials have become a near de facto standard for a majority of power-device applications used in global AI data centers.

We believe Sumitomo Bakelite can maintain its leadership position in the encapsulation market for three key reasons.

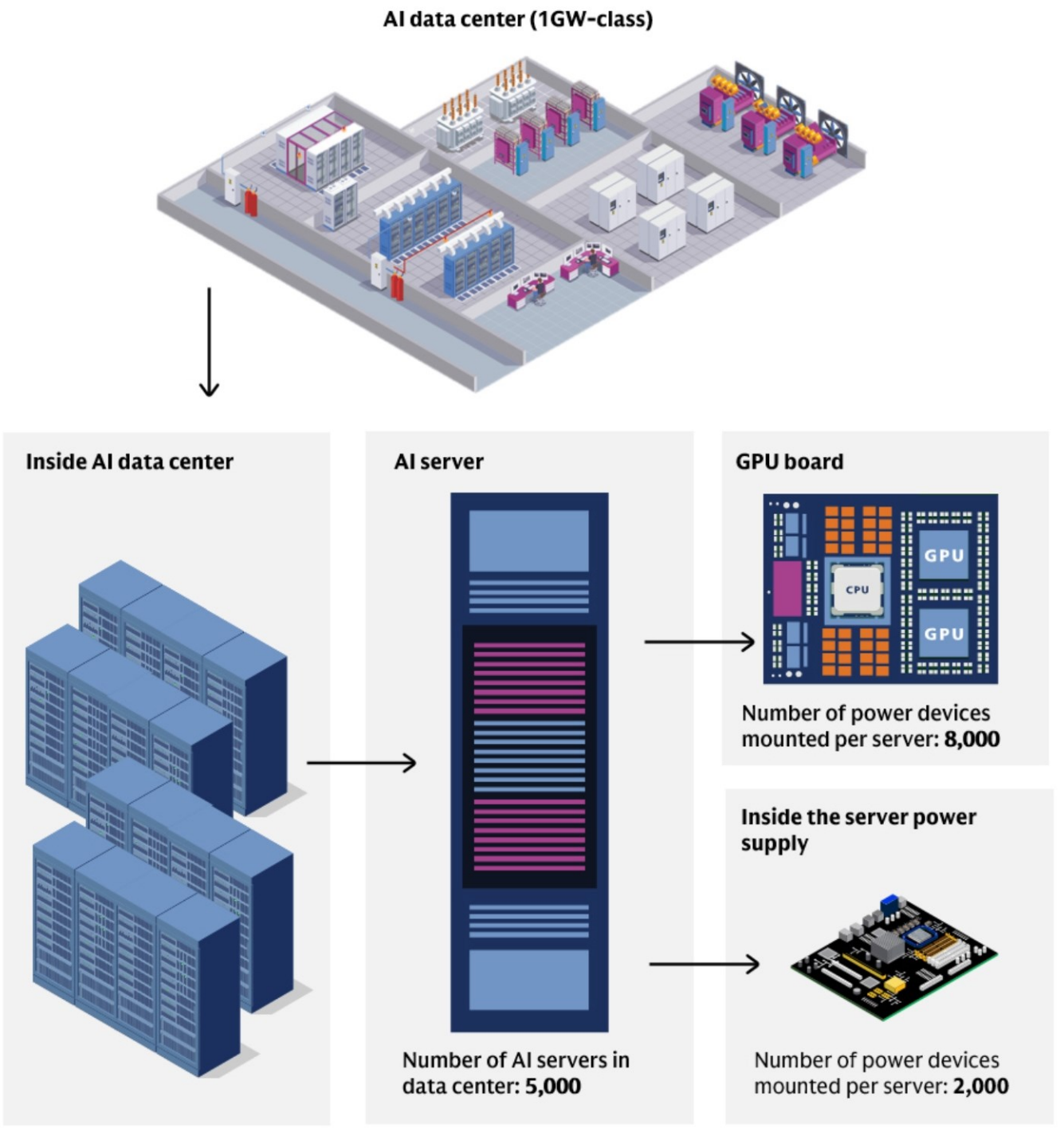
- First, the reliability requirements are highly demanding. AI-server power devices must continuously deliver large currents to GPUs while operating 24/7 under high utilization. Temperature fluctuations associated with AI training workloads place

severe demands on material reliability. EMC must provide durable adhesion, low-stress performance under thermal cycling, stable insulation, and consistent thermal conductivity even under moisture exposure. As power architectures migrate toward 800V-class systems, requirements for electrical insulation and ultra-low ionic contamination become even more stringent. Moreover, a 1GW AI data center can contain c.50 mn power devices. A single short-circuit event on a GPU board could trigger protection mechanisms that halt the board and potentially interrupt AI training jobs running across thousands of synchronized GPUs. Because EMC costs represent only a small fraction of total device costs, any savings achieved through lower-cost materials are insignificant relative to the potential losses associated with system failures. We believe this asymmetry favors established, highly qualified suppliers such as Sumitomo Bakelite.

- Second is technological complexity. Raising thermal conductivity requires ultra-high loadings of alumina, which has thermal conductivity several tens of times greater than silica. However, alumina is heavy, abrasive, and significantly increases viscosity. A formulation that combines the flowability needed for small, thin packages with the low-stress properties required to avoid cracking at high Tg represents a significant technological hurdle and cannot be developed overnight.
- Third is customer qualification and lengthy certification cycles. Power devices for AI servers are supplied primarily by global leaders such as Infineon, Monolithic Power Systems (MPS), Renesas, Texas Instruments, and ON Semiconductor. While assembly operations are frequently located in mainland China, materials qualification is typically conducted globally based on hyperscaler requirements.

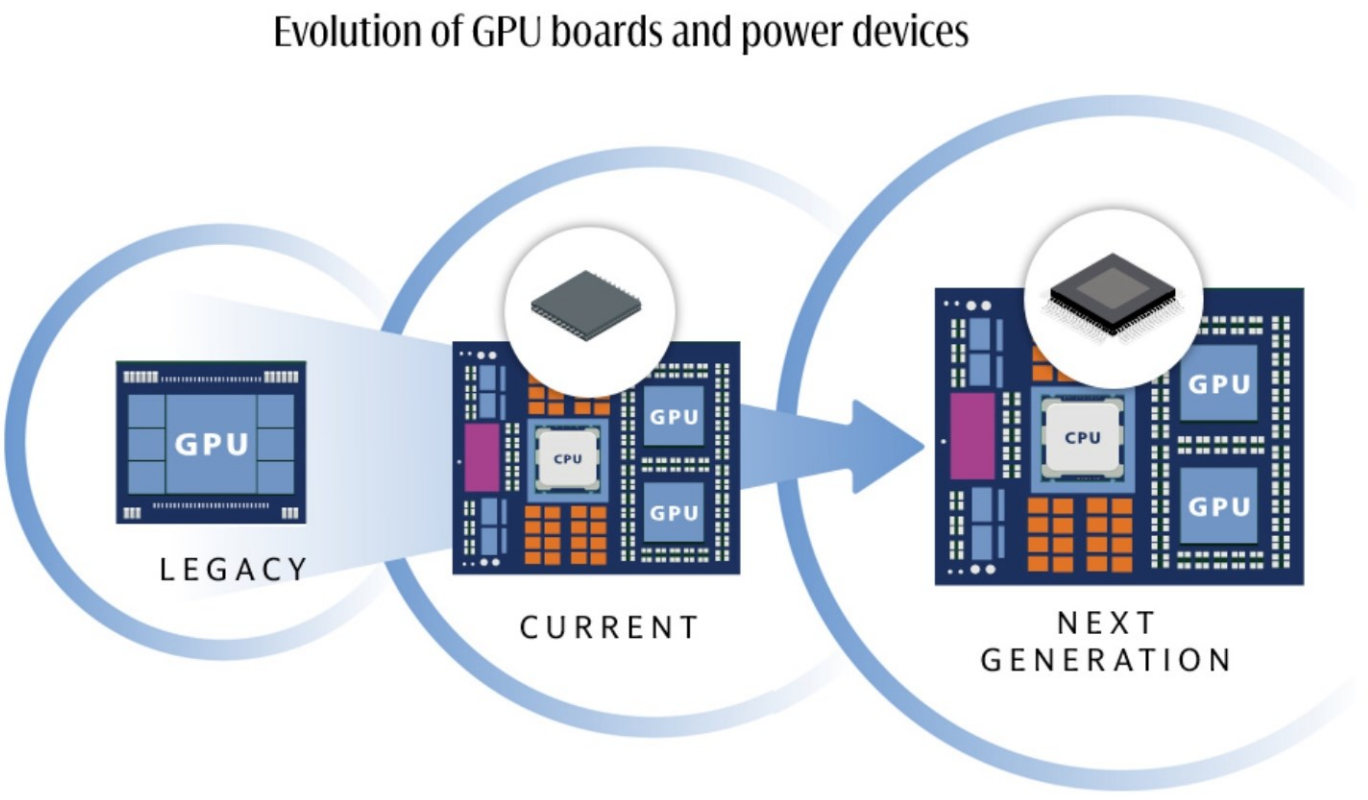
Based on our AI server demand assumptions of +86% YoY in 2026, +84% YoY in 2027, and +44% YoY in 2028 (value basis), and assuming the company maintains its current high market share while benefiting from an increasing mix of high-thermal-conductivity grades, we project an AI-server power-device encapsulation material revenue CAGR of c. 87% over FY3/26–FY3/29.

Exhibit 13: Inside an AI data center: Key infrastructure components



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 14: Rising demand for high-thermal-conductivity materials driven by GPU advancements



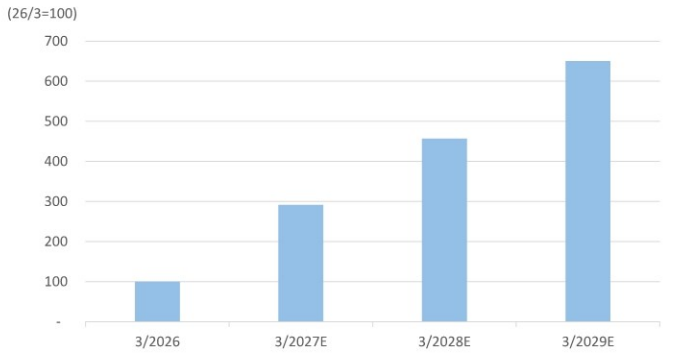
Number of power devices mounted and specifications

Power consumption	600W	3,000W	6,000W
Number of power devices	50	200	400
Specifications	Standard	High thermal conductivity	High thermal conductivity

The sharp increase in power consumption and device count is making heat dissipation a critical challenge.

Source: Company data, Goldman Sachs Global Investment Research

Exhibit 15: Sumitomo Bakelite: AI data center power encapsulation materials revenue (FY3/26=100, GSE)



Source: Company data, Goldman Sachs Global Investment Research

2) LEO satellites

Demand for advanced encapsulation materials in LEO satellite communications is being driven by two key structural characteristics of the satellite communication modules used in user terminals.

First, the receiving antenna employs a phased-array architecture consisting of several hundred to more than 1,500 antenna elements. Because each element requires its own RF front-end module, a single terminal can contain hundreds of RF chips. Moreover, communication with rapidly moving satellites using weak high-frequency signals requires minimizing electrical transmission losses. Conventional wire-bonded or substrate-based flip-chip structures suffer from relatively long signal paths and higher dielectric losses. As a result, Fan-In/Fan-Out packaging architectures, which form low-dielectric redistribution layers (RDLs) directly above the chip, are increasingly adopted (Exhibit 17). In these structures, interconnects are built on top of the encapsulated chip surface, making the planarity and warpage characteristics of the encapsulation material critical factors determining manufacturing yield.

There are two primary mass-production methods for Fan-In/Fan-Out packaging: Wafer-Level Packaging (WLP), which uses conventional 300 mm wafers, and Panel-Level Packaging (PLP), which utilizes large panels measuring c. 310 mm square-710 mm square (Exhibit 18). WLP relies on photolithography to form fine-pitch interconnects and typically uses liquid encapsulation materials. By contrast, PLP employs the semi-additive process (SAP), enabling lower-cost, high-volume manufacturing. Because communication applications generally require less demanding line widths than leading-edge logic semiconductors, PLP is particularly well suited for high-volume RF chip production. As encapsulation-material technology advances, we believe large-format PLP adoption will continue to expand.

The key technical challenge in scaling PLP to larger panel sizes is achieving uniform molding and controlling warpage across the entire panel surface. Liquid encapsulants tend to suffer from coating non-uniformity and increased warpage as panel sizes increase. By contrast, granular molding compounds offer a structural advantage. Their low-viscosity and low-stress material design allows them to be evenly distributed and molded across panels as large as 700 mm square-class formats.

According to management, Sumitomo Bakelite supplies granular molding compounds to a major semiconductor manufacturer producing modules for phased-array receiving antennas. Given limited players, we assume this customer is associated with STMicroelectronics' PLP-based RF module production. In addition, Sumitomo Bakelite announced in May 2026 that it had signed a long-term supply agreement with a major US aerospace company, which we believe could be related to the Starlink ecosystem. Per Sumitomo Bakelite, adoption of granular encapsulation materials for LEO applications began in FY3/26, with full-scale volume production expected to commence in the second half of FY3/27. The company expects FY3/27 sales of LEO-related granular encapsulation materials to roughly double year-on-year.

We expect the next phase of growth to come from receiver terminals, satellite terminals, and orbital data centers. Current adoption is concentrated in receiving-terminal antenna modules, where encapsulation-material consumption is estimated within the industry at several grams to several tens of grams per terminal. For receiver terminals, we forecast demand to increase from approximately 100 tonnes per year in FY3/27 to around 250 tonnes per year by FY3/29.

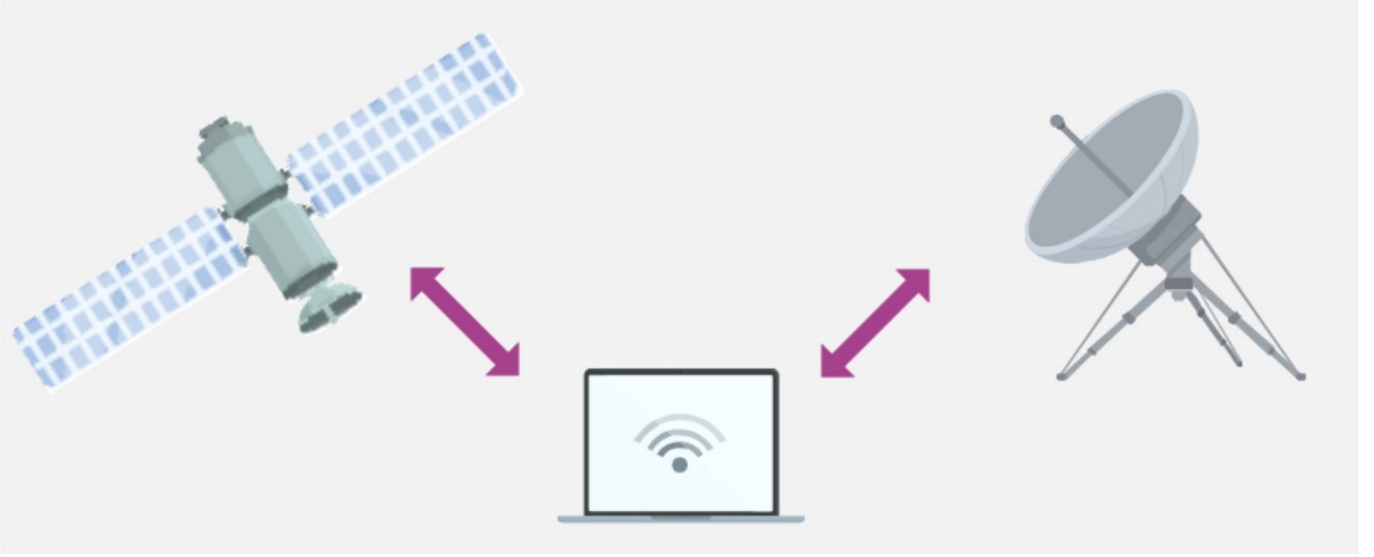
The next major opportunity lies in satellite-mounted communication and AI processors, where component counts can exceed 20,000 chips per satellite. Importantly, satellites introduce additional performance requirements not encountered in terrestrial applications. Traditional space-grade semiconductors have largely relied on hermetically sealed ceramic packages, with limited use of resin encapsulation. However, LEO satellite constellations, which are replaced on relatively short cycles, increasingly utilize commercial-style resin-encapsulated semiconductor packages.

As a result, encapsulation materials used in satellite applications must provide: a) Low outgassing characteristics to prevent volatile emissions in vacuum environments and 2) Low-stress performance and robust adhesion to withstand the dozen or more thermal cycles experienced each day as satellites transition between sunlight and darkness. Radiation tolerance, by contrast, is typically addressed primarily at the semiconductor-device level, and therefore the direct impact on encapsulation materials is likely limited.

We believe the company's estimate of several thousand tons per year of demand for advanced encapsulation materials reflects a long-term opportunity that extends beyond satellites themselves to include power-management systems for orbital AI data centers. This likely incorporates concepts such as SpaceX's proposed orbital computing infrastructure, which could ultimately comprise up to one million satellites and as much as 100GW of computational capacity.

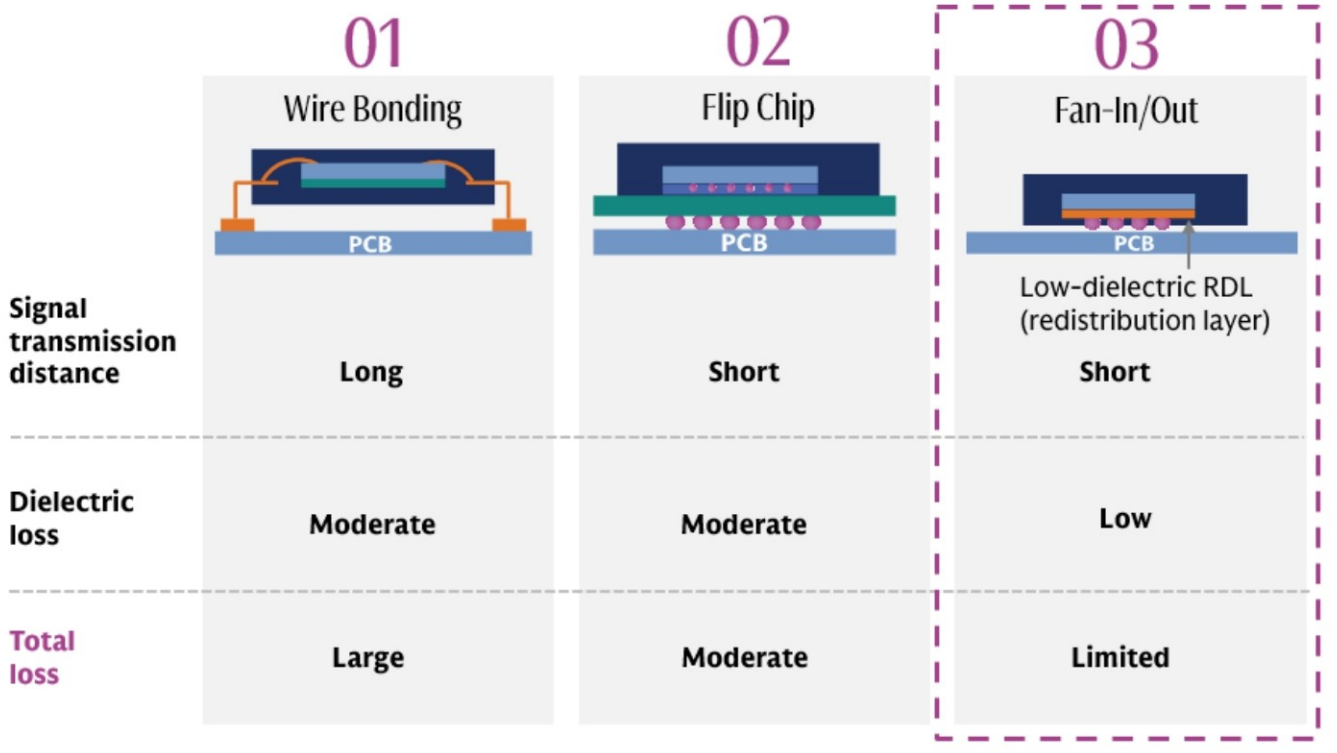
While the near-term earnings contribution remains limited, securing a position as the global standard for encapsulation materials used in PLP-based RF modules is strategically significant. Even if SpaceX proceeds with its planned in-house FO-PLP manufacturing facilities, material qualifications are typically carried forward once established, which should help sustain Sumitomo Bakelite's competitive advantage. We forecast revenue from space- and satellite-related applications to grow at a high CAGR of c. 118% between FY3/26 and FY3/29.

Exhibit 16: Communications satellites and receiving antennas



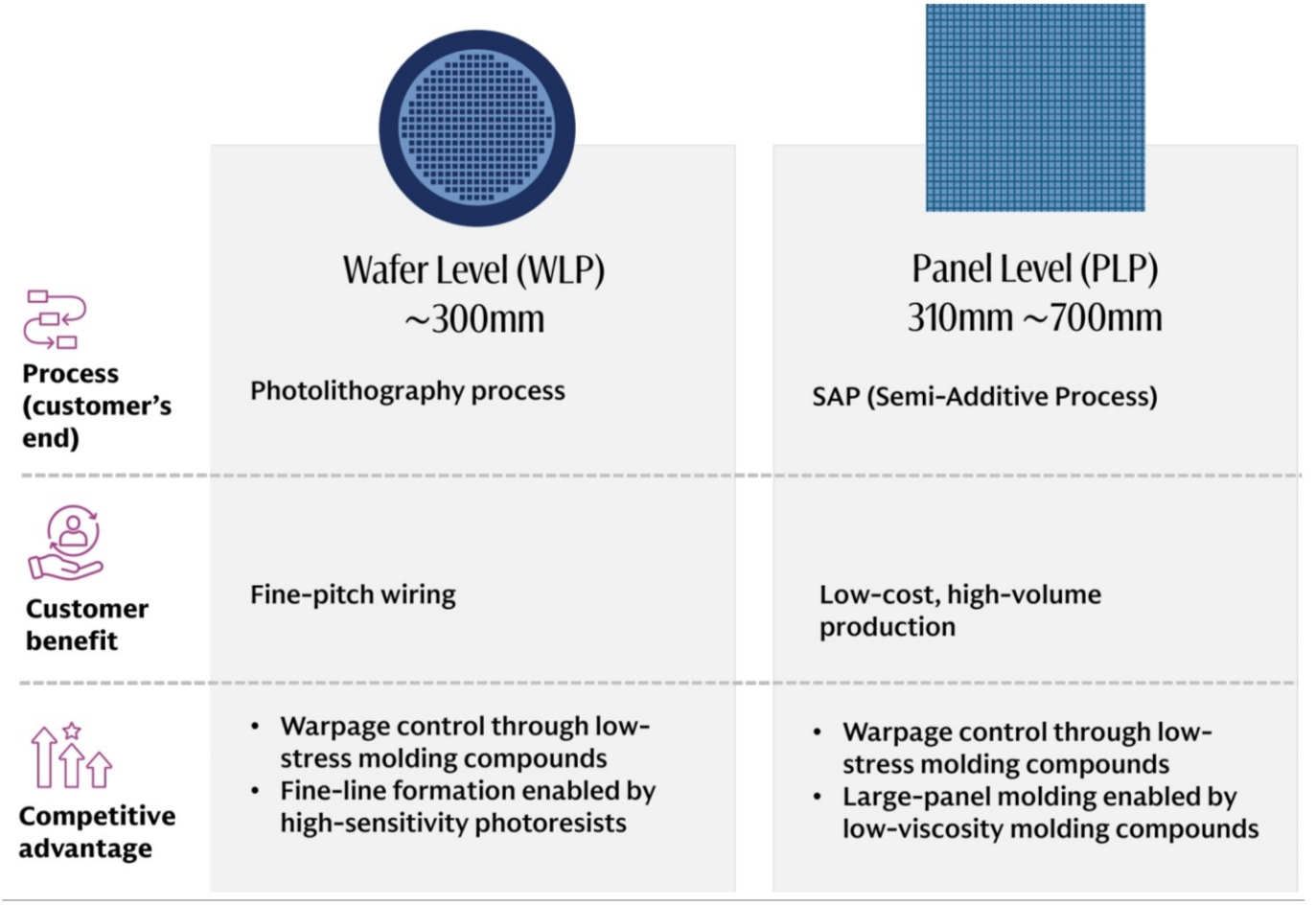
Source: Goldman Sachs Global Investment Research

Exhibit 17: Fan-out architecture minimizing transmission loss in satellite communications



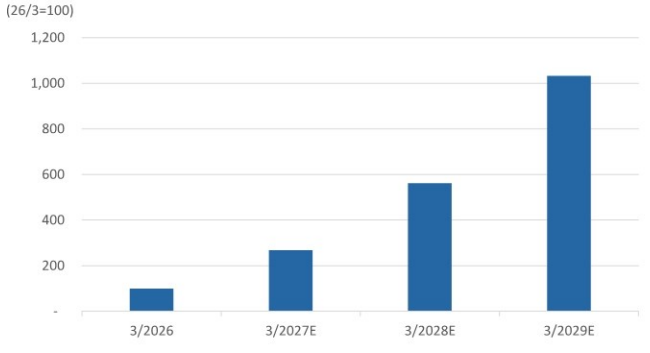
Source: Company data, Goldman Sachs Global Investment Research

Exhibit 18: Mass production process for Fan-In/Fan-Out RF modules



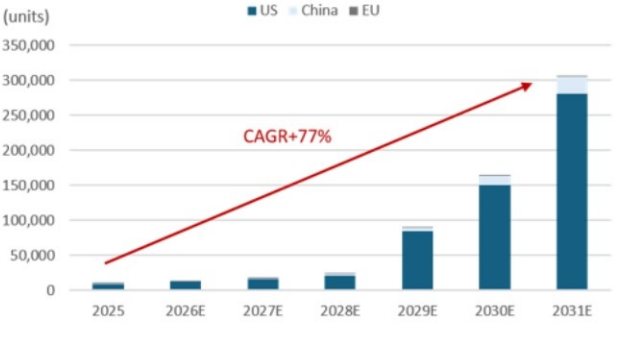
Source: Company data, Goldman Sachs Global Investment Research

Exhibit 19: Encapsulation materials revenue from LEO applications (FY3/26=100, our estimate)



Source: Goldman Sachs Global Investment Research

Exhibit 20: The global installed base of LEO satellites to reach 24k units by 2028E and 305k units by 2031E (GSE)



Source: Company data, Goldman Sachs Global Investment Research

3) Mobility

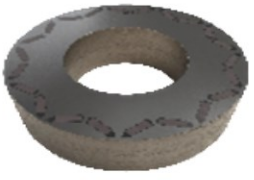
Revenue from the company's three strategic mobility products, namely motor magnet encapsulation materials, ECU/TCU encapsulation materials, and power-module encapsulation materials, nearly doubled between FY3/23 and FY3/26. We project a further c. 20% increase in FY3/27. For motor magnet encapsulation materials, usage per

vehicle is broadly similar in both HEVs and BEVs, as encapsulation materials are used to secure rotor magnets in either powertrain architecture. Consequently, the global shift toward vehicle electrification continues to serve as a key growth driver for the business.

ECU encapsulation materials are used to encapsulate entire A4-sized circuit boards for European Tier-1 suppliers. Sumitomo Bakelite has established a dominant market position by offering superior productivity through molds with significantly longer service lives than competing products, high mold-release performance, low thermal expansion and shrinkage, and strong customization capabilities.

Power-module encapsulation materials have expanded from Japanese and North American HEV programs into premium EV platforms in Europe and mainland China. In particular, for next-generation SiC power modules designed to operate at temperatures exceeding 200°C, the company developed the EME-G785 Series, which combines what the company describes as the world's highest-class glass-transition temperature (Tg 230°C) among solid epoxy molding compounds with proprietary low-stress characteristics. Mass production commenced in June 2026. Management positions the G785 Series as a strategic product for next-generation power electronics and is targeting JPY10 bn in annual sales by FY3/31.

Exhibit 21: Three strategic products (mobility)

Encapsulation materials for fixing motor magnets	Encapsulation materials for ECU/TCU	Encapsulation materials for power modules
		

Source: Company data

4) Physical AI

Humanoid robots are expected by the company to incorporate c. 100 small motors per unit in articulated joints. Assuming the market ultimately reaches annual shipments of 100 mn robots, this would translate into demand for roughly 10 bn motors per year. Sumitomo Bakelite is adapting its encapsulation technology originally developed for EV motor-magnet fixation to actuator applications such as robotic finger joints. According to the company, adoption has already begun at a major US customer. We estimate encapsulation-material consumption per humanoid robot at c. 0.5-2.0 kg per unit, suggesting a meaningful future opportunity. Management views humanoid robotics as a medium-term growth option beginning around 2028, when broader commercial deployment of robots is likely to accelerate.

Potential synergies from the Kyocera chemicals business acquisition

The acquisition of Kyocera's chemicals business, scheduled to close in October 2026, is priced at c.JPY30 bn. The acquired business generated c. JPY23.2 bn of revenue in FY3/26, implying an acquisition multiple of roughly 1.3x sales. Given Sumitomo Bakelite's

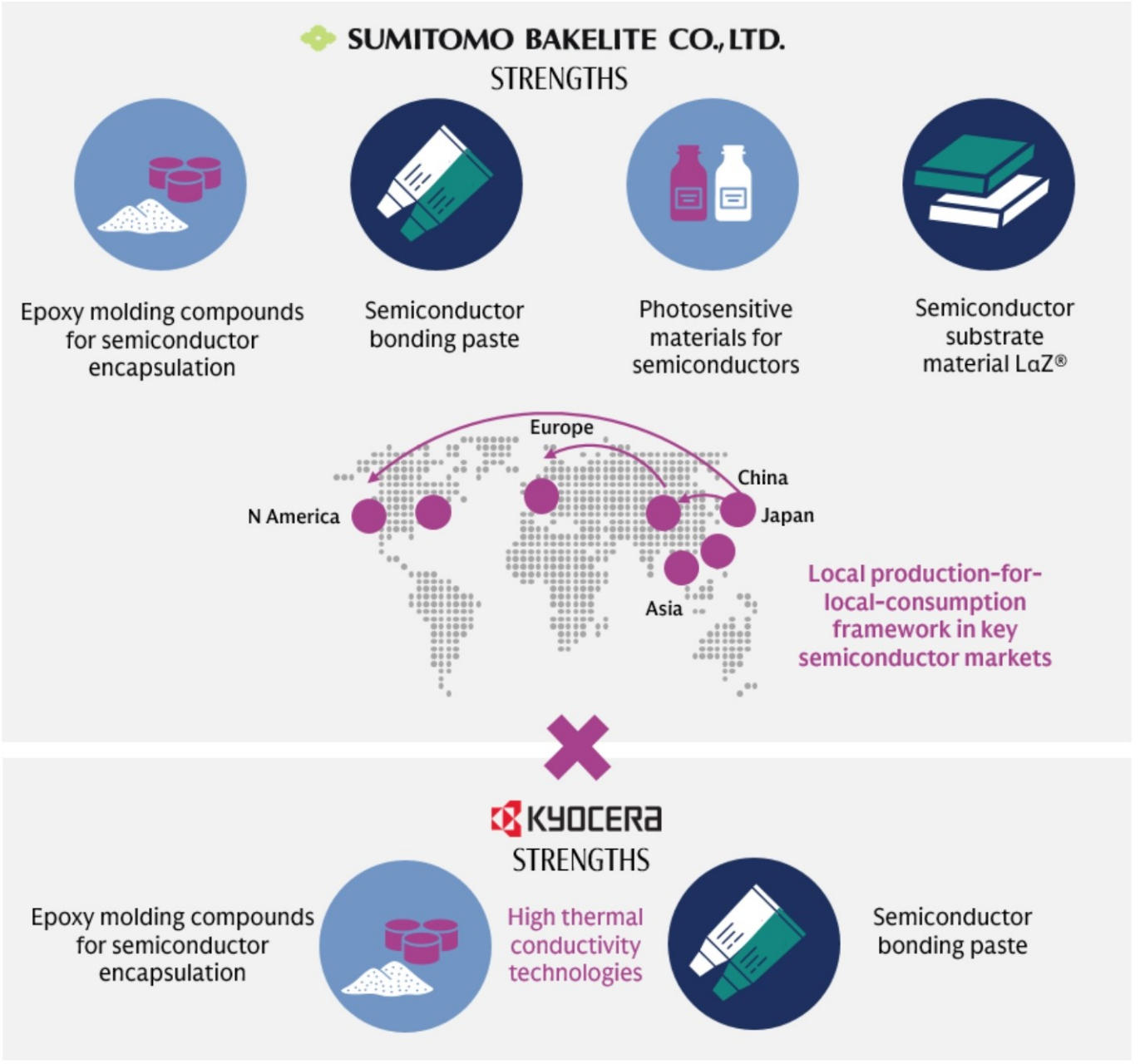
substantial net cash position of JPY104.7 bn (FY3/26), we believe the transaction can be funded entirely from internal resources, which would imply a limited financial burden. In contrast, the strategic merits are potentially significant. Approximately 70% of the acquired business is related to semiconductor materials, including semiconductor encapsulation materials (EMC) and bonding pastes, while the remaining 30% comprises industrial resins and high-performance plastics. The central strategic rationale lies in strengthening Sumitomo Bakelite's position in AI-data-center semiconductors, where demand for thermal-management solutions continues to increase rapidly.

The transaction also includes manufacturing facilities in Kawasaki, Moka, Koriyama, and Wuxi, suggesting potential for meaningful supply-side synergies, particularly through expanded capacity for advanced granular molding compounds. Following the acquisition, we calculate that Sumitomo Bakelite's global EMC market share will increase from just over 40% to more than 45%.

By integrating Kyocera's (formerly Toshiba Chemical's) high-thermal-conductivity technologies, Sumitomo Bakelite will be in a position to significantly broaden its portfolio of high-thermal-conductivity granular molding compounds, MUF materials, and conductive pastes across three key markets: advanced packaging, power devices, and memory. The memory segment in particular represents a meaningful opportunity, as Sumitomo Bakelite has historically had limited exposure to memory customers. By leveraging the former Kyocera's strong relationships with major US memory manufacturers, the company should be better positioned to offer high-thermal-conductivity MUF together with buffer coatings for DRAM BOC and FC-BGA applications in AI servers. Kyocera is also a leading supplier of silver sintering paste, which should further strengthen Sumitomo Bakelite's thermal-management offering for power and automotive applications.

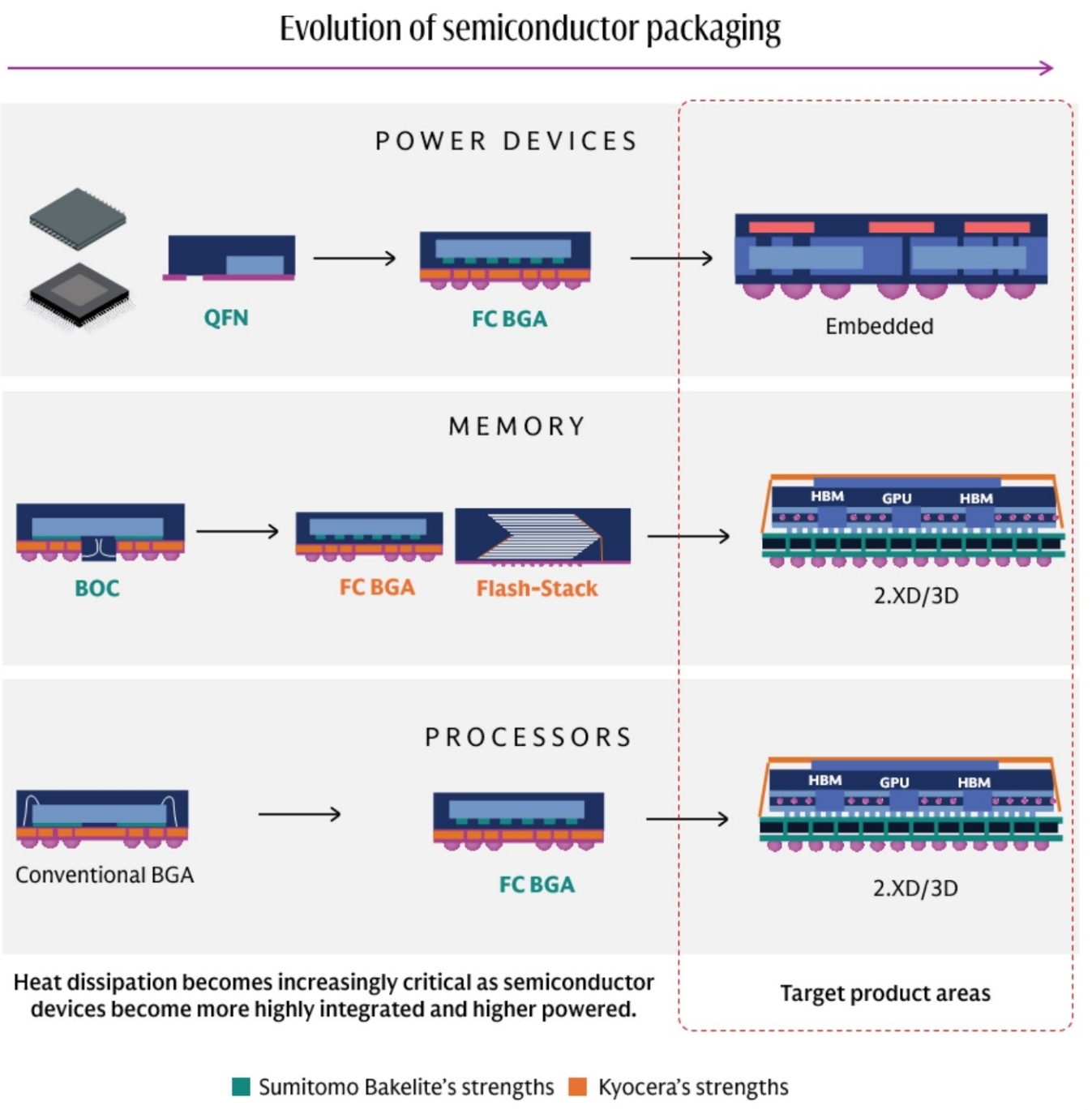
We estimate that the acquired business operates at profit margins exceeding those of Sumitomo Bakelite's existing Semiconductor Materials segment (20%). Even after accounting for purchase price allocation (PPA) amortization, we calculate that the transaction will make a meaningful earnings contribution, particularly from FY3/28 onward. From a synergy perspective, the company aims to combine Kyocera's strengths in BGA-related materials and thermal-management solutions with Sumitomo Bakelite's established positions in encapsulation and other materials. This should broaden the company's addressable market and enhance competitiveness in advanced packaging applications such as 2.5D and 3D packaging. For Sumitomo Bakelite, which historically has had limited commercial exposure to HBM encapsulation materials, acquiring relationships with memory customers provides a strategic commercial entry point into the rapidly growing HBM ecosystem.

Exhibit 22: Potential synergies from the proposed acquisition of Kyocera's chemicals business



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 23: Complementary strengths of Sumitomo Bakelite and Kyocera's chemicals business



Source: Company data, Goldman Sachs Global Investment Research

Photosensitive dielectric materials

SUMIRESIN EXCEL CRC, a positive-tone photosensitive wafer coating resin, was first mass-produced by Sumitomo Bakelite in 1997 as a positive-tone PBO-based material. It is used as both a buffer coat and an insulating layer for redistribution layers (RDLs). In addition to being used in DRAM devices supplied to Korea's two major memory manufacturers, we believe the material is also adopted in the uppermost layers of HBM packages. Going forward, demand should continue to grow steadily, driven primarily by memories used in AI data centers. The company is also expanding its presence in RDL

applications for Chinese power semiconductors, smartphone application processors, and edge AI devices.

Major competitors include HD Microsystems, Toray, and Asahi Kasei. As leading-edge RDL structures continue to migrate toward positive-tone polyimide-based materials for finer patterning requirements, management views SUMIRESIN EXCEL CRC as a stable profit contributor.

Die-bonding materials

Silver paste and silver sintering paste are used to attach semiconductor chips to lead frames and substrates. Demand is expanding, particularly for highly reliable products used in power management ICs, power devices, and automotive applications. The customer base is also broadening beyond its traditional concentration in Taiwan's consumer-electronics supply chain, with increasing penetration in mainland China and Southeast Asia. A key advantage for Sumitomo Bakelite is its ability to offer integrated material solutions, combining die-bonding materials with encapsulation compounds whose mechanical and thermal properties are optimized to work together. The acquisition of Kyocera's chemicals business should further strengthen this capability by adding silver sintering technologies to enhance the company's lineup of high-thermal-conductivity joining materials, including silver semi-sintering materials for 3D power modules.

We would note the impact of silver prices on this business. Since silver costs are relatively quickly reflected in selling prices, reported revenue tends to fluctuate more significantly than underlying shipment volumes. We estimate that die-bonding material sales more than doubled year-on-year in FY3/27 Q1; however, much of that increase likely reflected higher silver prices rather than pure volume growth. Given that silver prices have subsequently declined from peak levels, we expect revenue growth to normalize from Q2 onward.

LαZ

LαZ, a semiconductor package-substrate material, is currently used primarily in power inductors for high-end smartphones. However, demand is emerging for high-thermal-conductivity grades used in Wi-Fi base stations, AI server power devices, and embedded-MLCC substrates. The competitive barriers in AI-server CCL applications remain high, with established players such as Resonac maintaining strong positions. Nevertheless, if Sumitomo Bakelite can successfully expand sales of LαZ alongside its existing portfolio of encapsulation materials for AI server boards and die-bonding pastes, investor interest in the business could increase significantly.

High-Performance Plastics: Expanding earnings base mainly through aerospace and cycloolefin polymers

The High-Performance Plastics segment is centered on phenolic resins and phenolic molding compounds. The business portfolio comprises molding materials for automotive and industrial applications, resin products in North America, industrial phenolic resins, aircraft interior components, and newer products such as cycloolefin polymers for advanced semiconductors, thermally conductive insulating sheets for power modules, and other high-temperature-resistant materials.

The company is pursuing a dual-track strategy of enhancing margins in its existing businesses while shifting the product mix toward higher-value-added applications. Sales exposure to its designated strategic growth areas of electrified vehicles, semiconductors, and aerospace increased from 11% of segment sales in FY3/24 to 17% in FY3/26, with a target of more than 45% by FY3/31.

Within its existing businesses, the company continues to optimize its product portfolio, particularly in North America, while reorganizing manufacturing footprints across Japan and Asia. The company is also integrating phenolic molding compound production at Yamaroku Kasei Industry as part of its ongoing manufacturing optimization efforts.

The business areas we are focused on are outlined below.

Aircraft business

The aircraft business combines Sumitomo Bakelite's thermoset-material technologies with the aerospace industry channels acquired through the purchase of Vaupell around a decade ago. Under this structure, Sumitomo Bakelite supplies materials and intermediate products such as interior resins, aircraft interior panels, and cargo liners, while Vaupell manufactures finished aerospace components, including wing-unit components and air grilles.

According to the company, Boeing-related programs currently represent a large portion of sales, but the customer base is broadening. In July 2025, for example, the company began shipments of cargo liners for the new Airbus A350F freighter. We also expect the company's core aircraft interiors business to deliver high-single-digit (%) annual sales growth over the medium term, supported by rising maintenance, repair, and overhaul (MRO) demand for aircraft interior refurbishment. Combined with ongoing restructuring benefits, the business has the potential to become an increasingly important earnings contributor, in our view.

Cycloolefin Polymer (COPLUS)

COPLUS is a cycloolefin polymer that combines high heat resistance, optical transparency, low water absorption, and solvent solubility in a single material. Its glass-transition temperature (Tg) exceeds 250°C, substantially higher than that of conventional cycloolefin polymers (COPs), which generally have Tg values below 140°C. At the same time, COPLUS offers the transparency and solvent processability that polyimide materials lack, making it a unique specialty polymer.

While Zeon has a high share of the conventional cycloolefin polymer market, Sumitomo

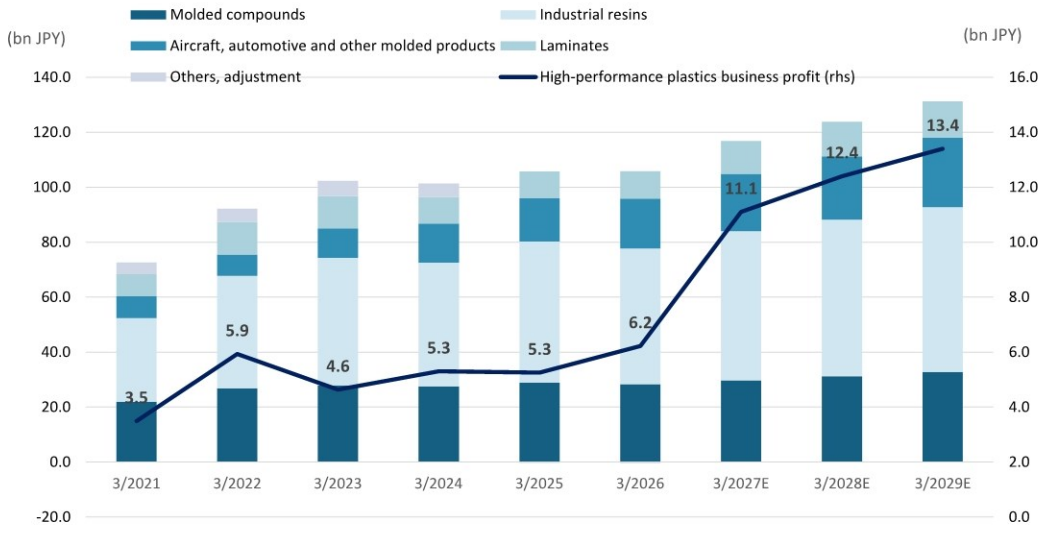
Bakelite is differentiating COPLUS through a proprietary norbornene-resin platform developed by subsidiary Promerus. This enables superior heat resistance and greater flexibility in polymer design. The company has said that sales into advanced semiconductor applications have already begun to ramp. Although the company has not disclosed customers or end uses, we believe the material may be used in optical-semiconductor applications, particularly lens-related components, given its unique combination of transparency and thermal stability. Looking ahead, we see potential for growth in high-speed photodiodes and optical transceiver applications used in AI data centers.

Exhibit 24: COPLUS offers superior heat resistance to conventional COPs while maintaining transparency and solvent solubility

Property	COPLUS	Conventional COP	Acrylic	Polyimide
Heat Resistance (Tg)	>250°C	<140°C	<100°C	>350°C
Transparency	○	○	○	×
Solvent Solubility	○	×	△	△

Source: Company data, Data compiled by Goldman Sachs Global Investment Research

Exhibit 25: High-Performance Plastics segment: Sales and business profits



Source: Company data, Goldman Sachs Global Investment Research

Quality of Life (QOL) Products: Stable earnings supported mainly by pharmaceutical and electronics-related products

The Quality of Life (QOL) Products segment consists of four businesses: films & sheets (FY3/26 sales weighting: 33%), industrial functional materials (37%), waterproofing materials (15%), and healthcare (15%).

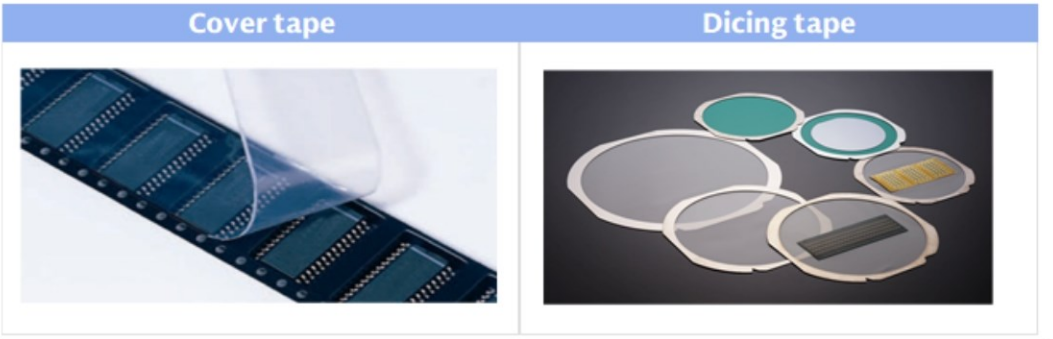
The company has actively expanded this segment through M&A. Notable transactions include the acquisition of Kawasumi Laboratories in 2020, Asahi Kasei Pax in 2024, and AGC's polycarbonate business in 2025. At the same time, management has aggressively pursued structural reforms. In medical devices, the company has shifted its portfolio toward minimally invasive treatment products and has exited the supply of dialysis blood tubing sets from its Thailand facility. As a result of these initiatives, segment business profits reached c.JPY12.9 billion in FY3/26, nearly doubling from FY3/21, while the business profit margin improved to c.12%, making the segment the company's second-largest stable earnings contributor after Semiconductor Materials. Importantly, a meaningful portion of this segment's end demand is linked to technology-related markets, including semiconductors and AI data centers. We therefore expect the segment to continue delivering stable, high growth. Key focus areas are outlined below.

Films & sheets business (cover tapes and dicing tapes)
Cover tapes are sealing films used to protect semiconductor chips and electronic components stored inside carrier tapes. These products must provide reliable sealing within extremely short cycle times while allowing stable peel-off during component mounting. They must also minimize contamination and defects so that components can be visually inspected through the tape. Sumitomo Bakelite has a high share in high-end applications through its expertise in sealant formulations and contamination control during film production. More recently, the company introduced a new product capable of reducing sealing time from 8 milliseconds to 5 milliseconds while maintaining required bonding strength, potentially improving customer productivity by c.20%.

Dicing tapes are adhesive tapes used during wafer singulation and chip pick-up processes. The company has developed strengths in sensor-related applications, including image sensors, by leveraging expertise in antistatic formulations that suppress electrostatic discharge during wafer cutting. Major competitors include Lintec and Nitto Denko.

As hybrid bonding is adopted more broadly, plasma dicing and laser full-cut processes that avoid contaminating bonding surfaces are becoming increasingly prevalent. Sumitomo Bakelite has developed substrate materials that resist wrinkling during plasma processing and are not damaged by laser cutting, as well as adhesive systems that remain stable during solvent cleaning. We think these technologies position the company to capture growth not only in its existing markets but also in next-generation HBM and 3D packaging applications.

Exhibit 26: Cover tape and dicing tape

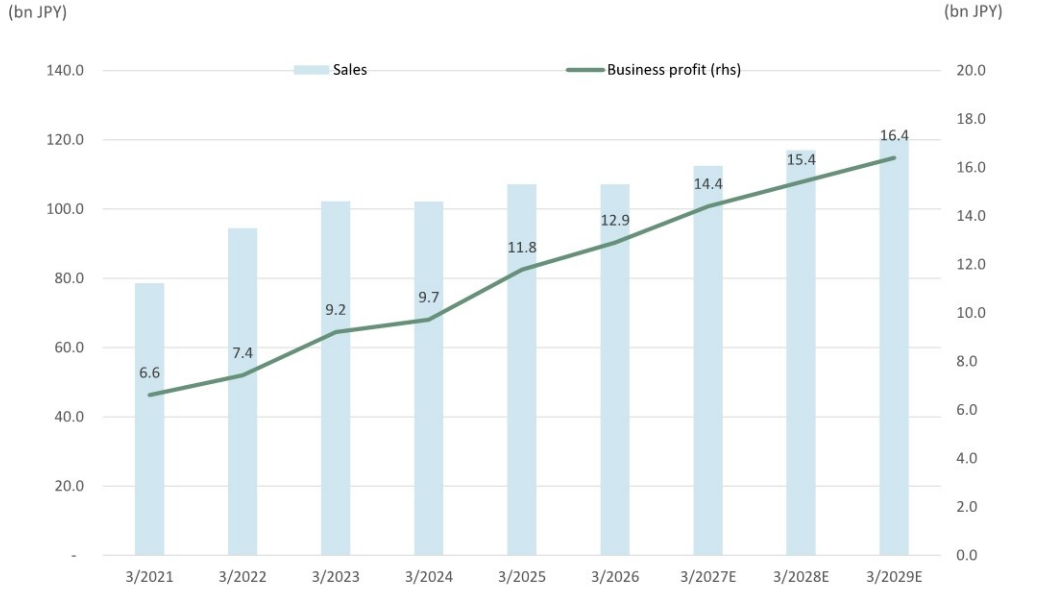


Source: Company data

Industrial functional materials business

This business is primarily centered on polycarbonate materials. General-purpose polycarbonate accounts for c.49% of segment sales (per FY3/27 guidance), but the company intends to reduce exposure to this relatively low-margin product to 42% by FY3/31, while increasing the contribution from higher-value-added applications as follows: a) Automotive optics: 12% → 18%, b) eyewear applications: 11% → 15%, and c) insulation materials: 8% → 12%. For data center applications, the company offers Eco-Sheet Polycarbonate, an industry-first, flame-retardant polycarbonate insulation sheet capable of supporting 800V-class electrical tracking requirements, as well as TwinCarbo, a hollow polycarbonate insulation panel. In automotive optics, Sumitomo Bakelite is strengthening its position in head-up display applications through ultra-low-retardation polycarbonate products acquired from AGC. In eyewear applications, the company is accelerating commercialization of Ecrare, an electronically dimmable sheet technology designed for smart glasses.

Exhibit 27: Quality of Life Products segment: Sales and business profits



Source: Company data, Goldman Sachs Global Investment Research

Valuation and Financial Position

Valuation

Our 12-month target price of JPY11,760 is derived using the relationship between EV/GCI and our forecast FY3/28 CROCI/WACC for the materials sector, assuming a cash return multiple of 0.7x. Given Sumitomo Bakelite's average EV/EBITDA valuation relative to the sector over the past three months, we apply neither a premium nor a discount to the sector multiple. Our target price is equivalent to an FY3/28E P/E of c.20X. We believe the company is entering a period of robust earnings growth, driven by expanding demand for semiconductor encapsulation materials in AI server and LEO-related applications, as well as potential synergies from the acquisition of Kyocera's chemicals business. Our FY3/27E-FY3/29E operating profit forecasts are 15%/26%/38% above Bloomberg consensus, respectively. While the stock has already risen 48% over the past six months, compared with a 7% gain for TOPIX (as of October 23), we believe meaningful re-rating potential remains. In our view, the market has yet to fully price in the company's earnings growth opportunities stemming from rising thermal-management demand in AI infrastructure and its growing exposure to satellite communications applications.

As a cross-check, we also conducted a sum-of-the-parts (SoTP) valuation analysis yielding an implied equity value of JPY12,080 per share, only 3% above our target price derived from relative valuation. The key assumptions for the SoTP analysis are as follows:

- **Semiconductor Materials:** We apply an FY3/28 EV/EBITDA multiple of 13x. This reflects our expectation for a robust 43% business profit CAGR between FY3/26 and FY3/29, driven primarily by the encapsulation materials (EMC) business, as well as valuation benchmarks observed among semiconductor and electronic-material peers.
- **High-Performance Plastics:** We apply an FY3/28 EV/EBITDA multiple of 6x. The valuation reflects ongoing earnings improvements from optimization of industrial resin and laminate manufacturing operations, together with continued aerospace-related demand.
- **Quality of Life (QOL) Products:** We apply an FY3/28 EV/EBITDA multiple of 8x. We believe this is justified by the segment's ability to generate c.JPY15 bn of annual profit through businesses such as semiconductor cover tapes and high-performance polycarbonate sheets, providing a stable and highly profitable earnings base.

Exhibit 28: SoTP valuation cross-check analysis

Sumitomo Bakelite (¥mn) 28/3 GSE	
Semiconductor materials EBITDA	55,100
Applied EV/EBITDA for semiconductor materials	13.0 x
EV from semiconductor materials	**716,300**
High-performance plastics EBITDA	18,700
Applied EV/EBITDA for high-performance plastics	6.0 x
EV from high-performance plastics	**112,200**
Quality of life products EBITDA	**20,400**
Applied EV/EBITDA for quality of life products	8.0 x
EV from quality of life products	**163,200**
Adjustment/others	(4,900)
Applied EV/EBITDA for adjustment/others	6.0 x
EV from adjustment/others	**(29,400)**
Total EV	**962,300**
Net debt (cash)	-100,932
Minority interest	3,459
Total equity value	1,059,773
Share count (mn)	88
Implied valuation at 28/3E (¥)	12,080

Source: Goldman Sachs Global Investment Research

Exhibit 29: Sumitomo Bakelite: PER historical trends

Source: Bloomberg

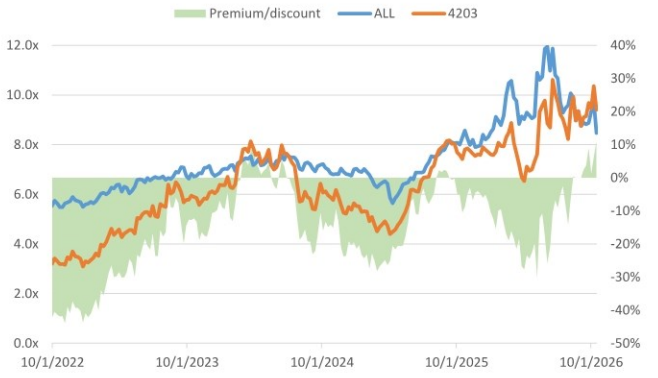
Exhibit 30: Sumitomo Bakelite: EV/EBITDA historical trends

Source: Bloomberg

Exhibit 31: Sumitomo Bakelite: PBR, ROE historical trends

Source: Bloomberg

Exhibit 32: Sumitomo Bakelite's premium/discount to the sector median EV/EBITDA multiple



Source: Bloomberg

Financial Position, Cash Flow, and Shareholder Returns

Financial position

Sumitomo Bakelite's financial foundation is strong, making it one of the most defensive companies in the materials sector with a highly robust balance sheet. As of FY3/26 year-end, the equity ratio attributable to owners of the parent stood at 71.7%, an increase of 2.1 pp from the previous year, with total shareholders' equity reaching c.JPY347.2 bn. Against cash and cash equivalents of JPY128.4 bn, interest-bearing debt, including both short- and long-term borrowings and commercial paper, totaled only JPY27.4 bn. As a result, the company maintains a substantial net cash position exceeding JPY100 bn, equivalent to a net D/E ratio of c.-0.26x.

Since FY3/21, the company has employed its proprietary SB-ROIC framework to rigorously monitor capital efficiency at the business-unit level. Corporate-wide ROIC improved steadily from 7.1% in FY3/24 to 8.9% in FY3/26, and management's long-term targets for 2030 include ROE of 10% and a business profit margin of 13%.

While the company plans to deploy capital toward the c.JPY30 bn acquisition of Kyocera's chemicals business and ongoing capacity expansion projects, we believe both the acquisition and growth investments can be comfortably funded through existing liquidity and internally generated cash flows, with minimal risk to financial stability.

Cash flow and capital allocation

Sumitomo Bakelite's cash-generation capability remains robust, supported by the strong growth and profitability of its Semiconductor Materials segment. Operating cash flow totaled JPY40.2 bn in FY3/24; JPY43.7 bn in FY3/25; and JPY35.0 bn in FY3/26. Free cash flow has remained consistently positive despite periodic increases in capital expenditures. FCF reached JPY28.1 bn in FY3/25 and JPY27.1 bn in FY3/26.

We would also highlight the attractive investment-return profile of its semiconductor encapsulation materials business. For EMC, capacity expansion can often be achieved through brownfield investments within existing sites, which we estimate require only around JPY5 bn per production line. Consequently, future growth investments should remain relatively capital efficient. Under the current Mid-Term Business Plan

(FY3/25-FY3/27), management's capital allocation has progressed largely in line with plan, while shareholder returns have significantly exceeded original targets:

- **Capital Expenditures and Growth Investments (Mid-Term Business Plan: JPY70 bn):** Capital spending has focused on rationalization of existing facilities, capacity expansion in mainland China and Southeast Asia, DX initiatives, and ERP upgrades. We expect cumulative capital expenditures over the three-year period to total c. JPY67 bn, remaining broadly within the original framework.
- **Strategic Investments (Mid-Term Business Plan: JPY50 bn):** Management has already committed c. JPY32.5 bn toward strategic acquisitions, including: AGC's polycarbonate business (c.JPY2.5 bn) and Kyocera's chemicals business (c.JPY30 bn). These transactions reflect a disciplined approach to portfolio transformation and expansion into higher-growth areas.
- **Shareholder Returns (Mid-Term Business Plan: JPY30 bn):** In FY3/25, the company executed a JPY20 bn share repurchase, supplemented by continuous dividend increases supported by stronger earnings performance. Including cumulative dividends of c.JPY30.7 bn, we expect total shareholder returns over the three-year period to reach c.JPY48.7 bn, substantially exceeding the original JPY30 bn target.

Shareholder return policy

Exceeding initial mid-term plan targets

Backed by a substantial cash position and strong free-cash-flow generation, Sumitomo Bakelite has become increasingly proactive in returning capital to shareholders. In its current Mid-Term Business Plan, the company raised its dividend payout guideline from 30% or more to c. 40% of consolidated net profit.

A notable feature of the company's capital-allocation strategy is its progress against the original shareholder-return target of JPY30 bn over the three-year Mid-Term Business Plan period. In addition to the JPY20 bn share repurchase executed in the plan's first year, we expect cumulative dividends to reach c.JPY30.7 bn, bringing total shareholder returns to roughly JPY48.7 bn over the three-year period. By funding M&A and capital expenditures in a disciplined manner within internally generated resources while returning excess cash to shareholders rather than allowing it to accumulate on the balance sheet, the company is on track to exceed its original shareholder-return commitment by a wide margin, achieving c. 160% of the initial target (JPY30 bn).

Dividend growth: a progressive dividend policy

On a post-stock-split basis, DPS has increased steadily:

- FY3/24: JPY75
- FY3/25: JPY95 (including a JPY5 70th-anniversary commemorative dividend; of JPY5 payout ratio 45%)
- FY3/26: JPY105 (payout ratio 33%)

The company plans a further dividend increase to JPY120 per share in FY3/27 (37% payout ratio, per company guidance). Supported by a de facto progressive dividend

policy under which dividends are not reduced, it offers shareholders a stable source of income.

Share repurchases and long-term capital policy
In FY3/25, the company repurchased c. JPY20 bn of its own shares (equivalent to roughly 5.9% of shares outstanding) through ToSTNeT-3 transactions. All repurchased shares were subsequently canceled on March 24, 2026. As part of its longer-term capital allocation strategy, management is considering introducing capital-linked shareholder return metrics, such as dividend on equity (DOE), in the next Mid-Term Business Plan, in addition to its current dividend payout ratio target. By actively managing its strong capital base (equity ratio of c. 71%), the company aims to lower its cost of capital while achieving its long-term ROE target of 10%.

History and corporate development

Sumitomo Bakelite is one of the pioneers of Japan's plastics industry. In 1911, prototype production of phenolic resin (Bakelite) began at the Shinagawa plant of Sankyo Goshi Kaisha (now Daiichi Sankyo), marking the first production of Bakelite in Japan. Nippon Bakelite, which inherited the business, was established in 1932. In 1955, it merged with Sumitomo Kakozai Kogyo to form today's Sumitomo Bakelite.

While rooted in its traditional synthetic resin business, Sumitomo Bakelite developed SUMIKON EME, an epoxy molding compound for semiconductor encapsulation, in 1968. Responding early to the shift of semiconductor assembly and test operations (OSATs) to Asia from the late 1980s onward, the company established local manufacturing bases across Singapore, Taiwan, and mainland China ahead of competitors. As a result, it has built a leading position in semiconductor encapsulation materials, commanding more than 40% global market share today, set to rise to c. 45% following the integration of Kyocera's chemicals business.

The company has also demonstrated a strong track record in M&A. Major acquisitions since the 2000s include:

- Occidental's phenolic-resins business (now Durez)
- Goodrich's electronic-materials research division (now Promerus)
- Vaupell (aircraft interior materials)
- Kawasumi Laboratories (now SB-Kawasumi Laboratories, medical devices)
- AGC's polycarbonate business (including automotive HUD materials)

In 2026, Sumitomo Bakelite agreed to acquire Kyocera's chemicals business for c.JPY30 bn, strengthening both its thermal-management technology portfolio and its access to major memory customers through strategic M&A.

Key Risks

- **AI server and advanced semiconductor demand volatility:** The Semiconductor Materials segment, the company's primary growth driver, benefits from expanding demand for AI-server GPUs, HBM, and advanced packaging applications through products such as EMC, die-bonding materials, and photosensitive dielectric materials. We forecast business profit in the Semiconductor Materials segment to

grow at a 43% CAGR between FY3/26 and FY3/29. However, if AI-server investment or advanced semiconductor demand were to fall below expectations, volume growth and product-mix improvement could slow, resulting in weaker earnings than we currently forecast.

- **Encapsulation materials competition and market share risk:** Although Sumitomo Bakelite holds a leading position in the EMC market, the competitive environment could change as existing competitors, including Chinese manufacturers, expand capacity and improve technology. We believe the company's technological advantage in high-end encapsulation materials for AI and high-performance computing applications remains substantial. However, if competing products were to gain traction more rapidly than expected in lower-end applications, pricing pressure and market-share shifts could negatively affect earnings.
- **FX:** The company derives a significant portion of its revenue from overseas markets and operates a global manufacturing and sales network spanning mainland China, Southeast Asia, North America, and other regions. As a result, foreign-exchange movements represent a meaningful earnings variable. Yen appreciation is generally unfavorable to profits. Our forecasts assume an exchange rate of JPY160/USD from FY3/27 Q2 onward.

M&A rank

We assign Sumitomo Bakelite an M&A Rank of 3 (Low Acquisition Probability) for two primary reasons:

- The company operates across multiple businesses with distinct end markets, including Semiconductor Materials, High-Performance Plastics, and Quality of Life Products. This diversified business portfolio could make potential acquirers hesitant, as it may require them to take on businesses outside their core strategic focus.
- The Semiconductor Materials business operates in areas that are increasingly subject to regulatory scrutiny and national-security considerations, particularly in cross-border transactions involving advanced semiconductor technologies. As a result, the regulatory environment surrounding potential acquisitions is likely to be relatively restrictive.

Exhibit 33: Sumitomo Bakelite: Financials

Profit model (¥bn)	3/26	3/27E	3/28E	3/29E
Revenue	319.9	376.6	424.0	465.4
Cost of goods sold	-219.8	-246.7	-274.0	-293.6
Gross profit	100.1	129.9	150.0	171.8
SG&A and other	-64.6	-74.2	-78.6	-86.4
Operating profit	35.5	55.7	71.4	85.4
Net interest income/(expense)	3.4	3.5	3.7	3.9
Equity in earnings of affiliates				
Net other nonoperating inc/(exp)	0.0	0.0	0.0	0.0
Recurring profit	38.8	59.2	75.1	89.3
Extraordinary income				
Extraordinary expense				
Pretax profit	38.8	59.2	75.1	89.3
Income tax	-10.5	-17.8	-22.5	-26.8
Minority interest	-0.4	-0.6	-0.7	-0.8
Net income	28.0	40.8	51.9	61.7
Capital expenditures (excl. leases)				
Capital expenditures (incl. leases)	-19.3	-30.0	-30.0	-25.0
Depreciation & amortization	-14.3	-15.5	-17.4	-18.7
EPS (basic) (¥)	319.5	465.4	591.7	703.2
EPS (fully diluted) (¥)	319.5	465.4	591.7	703.2
BVPS (¥)	3,996.6	4,431.5	4,951.6	5,544.7
DPS (¥)	105.0	160.0	220.0	280.0
Dividend payout ratio (%)	32.9	34.4	37.2	39.8

Year-on-year change (%)	3/26	3/27E	3/28E	3/29E
Revenue	5.0	17.7	12.6	9.8
Operating profit	43.1	57.0	28.2	19.6
Recurring profit	35.7	52.5	26.8	18.9
Net income	45.3	45.7	27.1	18.9
EPS (basic)	52.9	45.6	27.1	18.9
EPS (fully diluted)	52.9	45.6	27.1	18.9

Margins (%)	3/26	3/27E	3/28E	3/29E
Operating profit	11.1	14.8	16.8	18.3
EBITDA	15.5	18.9	20.9	22.4
Recurring profit	12.1	15.7	17.7	19.2
Net income	8.8	10.8	12.2	13.3

Cash flow statement (¥bn)	3/26	3/27E	3/28E	3/29E
Net income	28.0	40.8	51.9	61.7
D&A add-back	14.3	15.5	17.4	18.7
Minority interest add-back	0.4	0.6	0.7	0.8
Net (inc)/dec in working capital	-4.1	-13.0	-11.4	-9.5
Other operating cash flow	-3.5	0.0	-0.0	-0.0
Cash flow from operations	35.0	43.9	58.6	71.7
Capital expenditures (incl. leases)	-19.3	-30.0	-30.0	-25.0
Purchases of long-term securities				
Sales of long-term securities				
Other investing cash flow	11.3	0.0	0.0	0.0
Cash flow from investments	-7.9	-30.0	-30.0	-25.0
Dividends paid (common & preferred)	-9.2	-14.0	-19.3	-24.6
Incr./(decr.) in debt	-4.7	-7.3		
Common stock issuance	19.2	1.8	1.8	1.8
Other financing cash flow	-9.5	-1.8	-1.8	-1.8
Cash flow from financing	-4.2	-21.3	-19.3	-24.6
Total cash flow	22.8	-7.4	9.3	22.1

Balance sheet (¥bn)	3/26	3/27E	3/28E	3/29E
Cash & equivalents	128.4	121.0	130.3	152.5
Accounts receivable	57.9	68.1	76.7	84.2
Inventory	68.6	77.0	85.5	91.6
Other current assets	12.1	12.7	13.4	14.0
Total current assets	267.0	278.8	305.9	342.3
Net PP&E	141.4	155.9	168.5	174.8
Net intangibles	5.7	5.7	5.7	5.7
Total investments	0.0	0.0	0.0	0.0
Other long-term assets	70.1	80.2	91.9	105.3
Total assets	484.2	520.7	572.0	628.2
Accounts payable	46.1	51.8	57.5	61.6
Short-term debt	20.1	20.1	20.1	20.1
Other current liabilities	24.8	24.8	24.8	24.8
Total current liabilities	91.0	96.7	102.4	106.5
Long-term debt	7.3			
Other long-term liabilities	35.2	35.2	35.2	35.2
Total long-term liabilities	42.5	35.2	35.2	35.2
Total liabilities	133.5	131.9	137.6	141.7
Minority interest	3.5	4.1	4.8	5.6
Total common equity	347.2	384.7	429.7	480.9
Total liabilities & equity	484.2	520.7	572.0	628.2
Net debt	-101.1	-100.9	-110.3	-132.4

Ratios	3/26	3/27E	3/28E	3/29E
CROCI (%)	7.1	9.7	11.2	12.2
ROE (%)	8.7	11.0	12.6	13.4
ROA (%)	6.2	8.1	9.5	10.3
Net debt/equity (%)	-28.8	-26.0	-25.4	-27.2
Interest coverage ratio (X)	89.8	206.3	793.3	854.0

Valuation	3/26	3/27E	3/28E	3/29E
P/E (X)	14.6	16.7	13.2	11.1
EV/EBITDA (X)	11.8	8.2	6.5	5.3
EV/GCI (X)	0.6	1.0	0.9	0.8
P/B (X)	1.2	1.8	1.6	1.4
Dividend yield (%)	2.2	2.1	2.8	3.6
FCF yield (%)	2.3	2.0	4.2	6.8

Note: Last actual year may include reported and estimated data.

Source: Company data, Goldman Sachs Global Investment Research

Investment Thesis - Sumitomo Bakelite

Sumitomo Bakelite is a global leader in semiconductor back-end materials, holding a more than 40% global market share in semiconductor encapsulation materials (EMC; epoxy molding compounds) and leveraging proprietary thermal-management technologies developed through decades of resin formulation expertise. We expect strong growth in semiconductor materials driven by AI servers, advanced packaging, HBM-related applications, LEO satellite communications, and power semiconductors, supported by expanding sales of high-thermal-conductivity EMC, MUF, granular molding compounds, and die-bonding materials. We also expect additional earnings growth from disciplined M&A, particularly the acquisition of Kyocera's chemicals business, which strengthens the company's position in memory applications and thermal-management materials. The company shows strong earnings growth potential, supported by its high-growth Semiconductor Materials segment, stable profit expansion in its High-Performance Plastics and Quality of Life segments, and a strong net-cash balance

sheet that supports both growth investment and shareholder returns. We are Buy rated.

Price Target Risks and Methodology - Sumitomo Bakelite

Our 12-month target price of JPY11,760 is derived from the correlation between materials sector EV/GCI and FY3/28E CROCI/WACC (cash return multiple 0.7X). Key risks: (1) weaker-than-expected AI server and advanced semiconductor demand, (2) intensifying competition and market-share erosion in encapsulation materials (EMC), and (3) yen appreciation against USD.

Disclosure Appendix

Reg AC

We, Atsushi Ikeda and Yuri Izumikawa, hereby certify that all of the views expressed in this report accurately reflect our personal views about the subject company or companies and its or their securities. We also certify that no part of our compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: Atsushi Ikeda Goldman Sachs Japan Co., Ltd., Yuri Izumikawa Goldman Sachs Japan Co., Ltd..

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

GS Factor Profile

The Goldman Sachs Factor Profile provides investment context for a stock by comparing key attributes to the market (i.e. our universe of rated stocks) and its sector peers. The four key attributes depicted are: Growth, Financial Returns, Multiple (e.g. valuation) and Integrated (a composite of Growth, Financial Returns and Multiple). Growth, Financial Returns and Multiple are calculated by using normalized ranks for specific metrics for each stock. The normalized ranks for the metrics are then averaged and converted into percentiles for the relevant attribute. The precise calculation of each metric may vary depending on the fiscal year, industry and region, but the standard approach is as follows:

Growth is based on a stock's forward-looking sales growth, EBITDA growth and EPS growth (for financial stocks, only EPS and sales growth), with a higher percentile indicating a higher growth company. **Financial Returns** is based on a stock's forward-looking ROE, ROCE and CROCI (for financial stocks, only ROE), with a higher percentile indicating a company with higher financial returns. **Multiple** is based on a stock's forward-looking P/E, P/B, price/dividend (P/D), EV/EBITDA, EV/FCF and EV/Debt Adjusted Cash Flow (DACF) (for financial stocks, only P/E, P/B and P/D), with a higher percentile indicating a stock trading at a higher multiple. The **Integrated** percentile is calculated as the average of the Growth percentile, Financial Returns percentile and (100% - Multiple percentile).

Financial Returns and Multiple use the Goldman Sachs analyst forecasts at the fiscal year-end at least three quarters in the future. Growth uses inputs for the fiscal year at least seven quarters in the future compared with the year at least three quarters in the future (on a per-share basis for all metrics).

For a more detailed description of how we calculate the GS Factor Profile, please contact your GS representative.

M&A Rank

Across our global coverage, we examine stocks using an M&A framework, considering both qualitative factors and quantitative factors (which may vary across sectors and regions) to incorporate the potential that certain companies could be acquired. We then assign a M&A rank as a means of scoring companies under our rated coverage from 1 to 3, with 1 representing high (30%-50%) probability of the company becoming an acquisition target, 2 representing medium (15%-30%) probability and 3 representing low (0%-15%) probability. For companies ranked 1 or 2, in line with our standard departmental guidelines we incorporate an M&A component into our target price. M&A rank of 3 is considered immaterial and therefore does not factor into our price target, and may or may not be discussed in research.

Quantum

Quantum is Goldman Sachs' proprietary database providing access to detailed financial statement histories, forecasts and ratios. It can be used for in-depth analysis of a single company, or to make comparisons between companies in different sectors and markets.

Disclosures

The rating(s) for Sumitomo Bakelite Co. is/are relative to the other companies in its/their coverage universe: Asahi Kasei, Daicel, Denka, Fujimi Inc., JX Advanced Metals Corp., Kuraray, Mitsubishi Gas Chemical, Mitsui Chemicals Inc., Mitsui Kinzoku Co., Nippon Paint Holdings, Nitto Boseki Co, Resonac Holdings, SUMCO, Shin-Etsu Chemical, Sumitomo Bakelite Co., Tokyo Ohka Kogyo, Toray Industries, Tri Chemical Laboratories Inc., Zeon

Company-specific regulatory disclosures

The following disclosures relate to relationships between The Goldman Sachs Group, Inc. (with its affiliates, "Goldman Sachs") and companies covered by Goldman Sachs Global Investment Research and referred to in this research.

Goldman Sachs beneficially owned 1% or more of common equity (excluding positions managed by affiliates and business units not required to be aggregated under US securities law) as of the month end preceding this report: Sumitomo Bakelite Co. (¥7,839)

Goldman Sachs had a non-securities services client relationship during the past 12 months with: Sumitomo Bakelite Co. (¥7,839)

Distribution of ratings/investment banking relationships

Goldman Sachs Investment Research global Equity coverage universe

	Rating Distribution			Investment Banking Relationships		
	Buy	Hold	Sell	Buy	Hold	Sell
Global	50%	34%	16%	65%	58%	43%

As of October 1, 2026, Goldman Sachs Global Investment Research had investment ratings on 3,094 equity securities. Goldman Sachs assigns stocks as Buys and Sells on various regional Investment Lists; stocks not so assigned are deemed Neutral. Such assignments equate to Buy, Hold and Sell for the purposes of the above disclosure required by the FINRA Rules. See 'Ratings, Coverage universe and related definitions' below. The Investment Banking Relationships chart reflects the percentage of subject companies within each rating category for whom Goldman Sachs has provided investment banking services within the previous twelve months.

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or

co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; registered market maker. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at https://www.gs.com/worldwide/brazil/area/gir/index.html. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details, a copy of the annual compliance audit report and other relevant information and disclosures can be found at this link: https://www.goldmansachs.com/worldwide/india/research-analyst. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom's departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at https://www.gs.com/disclosures/europeanpolicy.html which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Ratings, coverage universe and related definitions

Buy (B), Neutral (N), Sell (S) Analysts recommend stocks as Buys or Sells for inclusion on various regional Investment Lists. Being assigned a Buy or Sell on an Investment List is determined by a stock's total return potential relative to its coverage universe. Any stock not assigned as a Buy or a Sell on an Investment List with an active rating (i.e., a stock that is not Rating Suspended, Not Rated, Early-Stage Biotech, Coverage Suspended or Not Covered), is deemed Neutral. Each region manages Regional Conviction Lists, which are selected from Buy rated stocks on the respective region's Investment Lists

and represent investment recommendations focused on the size of the total return potential and/or the likelihood of the realization of the return across their respective areas of coverage. The addition or removal of stocks from such Conviction Lists are managed by the Investment Review Committee or other designated committee in each respective region and do not represent a change in the analysts' investment rating for such stocks.

Total return potential represents the upside or downside differential between the current share price and the price target, including all paid or anticipated dividends, expected during the time horizon associated with the price target. Price targets are required for all covered stocks. The total return potential, price target and associated time horizon are stated in each report adding or reiterating an Investment List membership.

Coverage Universe: A list of all stocks in each coverage universe is available by primary analyst, stock and coverage universe at https://www.gs.com/research/hedge.html.

Not Rated (NR). The investment rating, target price and earnings estimates (where relevant) are removed pursuant to Goldman Sachs policy when Goldman Sachs is acting in an advisory capacity in a merger or in a strategic transaction involving this company, when there are legal, regulatory or policy constraints due to Goldman Sachs' involvement in a transaction, and in certain other circumstances. **Early-Stage Biotech (ES).** An investment rating and a target price are not assigned pursuant to Goldman Sachs policy when this company has neither a drug, treatment or medical device that has passed a Phase II clinical trial nor a license to distribute a post-Phase II drug, treatment or medical device. **Rating Suspended (RS).** Goldman Sachs Research has suspended the investment rating and price target for this stock, because there is not a sufficient fundamental basis for determining an investment rating or target price. The previous investment rating and target price, if any, are no longer in effect for this stock and should not be relied upon. **Coverage Suspended (CS).** Goldman Sachs has suspended coverage of this company. **Not Covered (NC).** Goldman Sachs does not cover this company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (https://www.sipc.org).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

The analysts named in this report may have from time to time discussed with our clients, including Goldman Sachs salespersons and traders, or may discuss in this report, trading strategies that reference catalysts or events that may have a near-term impact on the market price of the equity securities discussed in this report, which impact may be directionally counter to the analyst's published price target expectations for such stocks. Any such trading strategies are distinct from and do not affect the analyst's fundamental equity rating for such stocks, which rating reflects a stock's return potential relative to its coverage universe as described herein.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, make a market or act as liquidity provider in, and buy or sell, any securities, derivatives, or other financial instruments referred to in this research (including any financial instruments issued by the companies referred to herein), unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at https://www.theocc.com/about/publications/character-risks.jsp and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction

costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g., marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to https://research.gs.com.

Disclosure information is also available at https://www.gs.com/research/hedge.html or from Research Compliance, 200 West Street, New York, NY 10282.

© 2026 Goldman Sachs.

You are permitted to store, display, analyze, modify, reformat, and print the information made available to you via this service only for your own use. You may not resell or reverse engineer this information to calculate or develop any index for disclosure and/or marketing or create any other derivative works or commercial product(s), data or offering(s) without the express written consent of Goldman Sachs. You are not permitted to publish, transmit, or otherwise reproduce this information, in whole or in part, in any format to any third party without the express written consent of Goldman Sachs. You may not use, extract, download or retrieve this information, in whole or in part, to train, finetune, adjust, influence the weights or behavior of, or otherwise develop or improve a machine learning or artificial intelligence model or system, or provide or reproduce this information, in whole or in part, as a prompt or input to any such system without the express written consent of Goldman Sachs. Furthermore, you may not bulk download this information, or employ any robot, spider, crawler, scraper, automated agent, or other electronic means to scrape, harvest, extract, or retrieve this information, without the express written consent of Goldman Sachs. A breach of the above terms may result in termination of your right to access this service.